Virtual Reality (2026) 30:139
https://doi.org/10.1007/s10055-026-01388-1

SI.: CONNECTED REALITIES: EXPLORING COLLABORATION AND SOCIAL INTERACTION IN EXTENDED REALITY

Harassment in social virtual reality: frequency, harmfulness, and governance responses

Qijia Chen[1] · Andrea Bellucci[2] · Qunfang Wu[3] · Giulio Jacucci[1]

Received: 20 November 2025 / Accepted: 24 April 2026 / Published online: 19 May 2026
© The Author(s) 2026

Abstract
Harassment in social Virtual Reality (VR) poses distinctive risks as embodied and immersive interactions heighten the impact of harmful encounters. We surveyed 215 adult Social VR users on the frequency, perceived harmfulness, and preferred governance responses to verbal, physical, and environmental harassment. Verbal harassment was most frequent and broadly harmful. Physical harassment, though less frequent, elicited the strongest polarization: more users than in other categories rated it either not harmful at all or extremely harmful, highlighting how embodiment magnifies divergent interpretations of harassment. Gender differences emerged not in frequency but in perceived harm and governance preferences, with women consistently rating harassment as more harmful and strongly endorsing all responses. Across responses, bans were most supported, yet participants also expressed strong support for legal action. These results challenge prior online harassment research on both frequency and governance preferences, highlighting how embodiment and identity masking reshape risks and responses in social VR.

Keywords Harassment · Virtual reality · Social VR · Governance

1 Introduction

Social Virtual Reality (VR) platforms have become popular venues for real-time and embodied interaction, where users engage as avatars in immersive 3D environments (Piitulainen et al. 2022). These platforms combine avatar embodiment, shared space, and spatialized audio, affording interaction experiences that approximate physical co-presence (Mcveigh-Schultz et al. 2018). As social VR matures, scholars have documented the development of distinct cultures and norms (Chen et al. 2025; Mcveigh-Schultz et al. 2019; Maloney et al. 2021; Sabri et al. 2023), alongside new forms of conflict, disruption, and harassment (Blackwell et al. 2019; Sabri et al. 2023).

While harassment in online spaces is not new, and has long challenged designers and researchers across online forums, digital games, and social media platforms (Bruckman et al. 1994; Blackwell et al. 2017), the dynamics of harm shift considerably in immersive environments. In social VR, harassment is not limited to speech (or text) but may also involve spatial intrusion, unwanted physical gestures, or environmental manipulation[1] (Blackwell et al. 2019; Freeman et al. 2022; Chen et al. 2024; Freeman and Maloney 2021). The embodied and synchronous nature of these interactions may amplify emotional impact and reshape how harm is enacted and perceived (Weerasinghe et al. 2025).

✉ Qijia Chen
qijia.chen@helsinki.fi

Andrea Bellucci
abellucc@inf.uc3m.es

Qunfang Wu
qunfang.wu@fsu.edu

Giulio Jacucci
giulio.jacucci@helsinki.fi

[1] Department of Computer Science, University of Helsinki, Helsinki, Finland

[2] Department of Computer Science and Engineering, Universidad Carlos III de Madrid, Leganés, Spain

[3] School of Information, Florida State University, Tallahassee, FL, USA

[1] Environmental harassment refers to harassment enacted by manipulating the shared virtual environment (e.g., displaying explicit or otherwise offensive content on shared screens, placing harassing objects or imagery into a shared space, or altering the space in ways that target a user and make the environment feel hostile) (Blackwell et al. 2019).

Recent studies highlight a wide range of such harms (Blackwell et al. 2019; Freeman et al. 2022; Weerasinghe et al. 2025), including verbal abuse, physically intrusive behaviors like avatar groping, and environmental disruptions such as displaying toxic graphic content or throwing virtual objects. These incidents are often described as emotionally intense and difficult to dismiss (Blackwell et al. 2019). Their impact is shaped by a combination of perceptual realism and weak normative boundaries (Schulenberg et al. 2024; Blackwell et al. 2019). Prior work shows that women and marginalized groups are disproportionately targeted in online harassment and that women report heightened emotional distress even when exposure levels are comparable to men (Vogels 2021). Yet it remains unclear how these patterns translate to social VR, the distinct immersive online environment where identity expression is often fluid. Understanding not just the frequency of harassment but the perceived harm across demographic groups is essential to developing safety interventions that are equitable and responsive.

Platform responses to harassment in social VR remain dominated by punitive governance mechanisms such as bans (Kou 2021; Ma et al. 2023). While these measures provide immediate protection, critics argue that they do little to address deeper questions of justice, repair, and accountability (Schoenebeck et al. 2021). Alternative responses, such as apologies or compensation, have been proposed to complement bans by acknowledging harm and restoring user agency (Schoenebeck et al. 2023). Yet little is known about how users in immersive contexts value different remedies, or how these preferences vary by identity. To address these gaps, we conducted a survey with 215 adult social VR users. The study investigates both how different forms of harassment (verbal, physical, and environmental) are experienced in terms of frequency, harmfulness, and how users evaluate a range of governance responses, including banning, monetary compensation, apology, legal action, and identity exposure. We further analyze how these judgments vary across demographic groups, with particular attention to gender-based differences.

Our contributions to the literature are as follows: (1) We distinguish verbal, physical, and environmental harassment by analyzing their frequency and perceived harm, offering a more nuanced understanding of risk in immersive contexts. (2) We show that, unlike prior online harassment literature, harassment exposure frequency does not significantly differ by gender. However, clear gender differences emerge in perceived harm, particularly for physical harassment. This highlights critical implications for gender-sensitive design and policy in immersive environments. (3) We analyze user preferences for five governance responses, showing how expectations vary by harassment type and gender. Notably, strong support for legal action in cases of physical harassment raises questions about the sufficiency of platform governance alone. (4) We translate these findings into design insights to guide future safety tools and policies that are responsive to gendered experiences and diverse risks in immersive environments.

2 Related works

This section reviews prior work on online harassment, with a focus on justice models in platform governance. It then examines how social VR raises challenges in online harm.

2.1 Online harassment and its impact

Online harassment is a widespread and persistent issue across digital platforms, encompassing a broad spectrum of behaviors,including insults, threats, impersonation, unwanted contact, and doxxing.[2] These behaviors are mediated by digital technologies and often intensified by platform affordances such as anonymity, virality, and algorithmic amplification (Blackwell et al. 2017). Research across social media, forums, and online games consistently documents the high prevalence of harassment (Ybarra and Mitchell 2007; Thomas et al. 2021), its serious psychological and social impacts (Stevens et al. 2021; Ybarra 2004; Bruckman et al. 1994), and ongoing efforts to mitigate it through interventions like content removal, account bans, and automated detection (Kou 2021; Ma et al. 2023).

The psychological toll of harassment is well documented and includes anxiety, stress, fear, humiliation, and illness (Pittaro 2007; Jiang et al. 2021). These harms can also suppress user expression: for example, 27% of U.S. internet users report self-censoring due to fear of being harassed online (Celuch et al. 2024; Citron 2017). Harassment does not affect all users equally. Groups such as women, LGBTQ+ individuals, people of color, and disabled users experience disproportionate harm (Thomas et al. 2021), particularly at the intersection of these identities (Lenhart et al. 2016; Sambasivan et al. 2019; Sultana et al. 2018). Additionally, women tend to report higher perceived harm across harassment types compared to men (Im et al. 2022; Vogels 2021).

Researchers have increasingly focused not just on the occurrence of harassment, but also on how different forms of harm are perceived. Schoenebeck et al. (2023) found that adult social media users rated non-consensual photo

[2] Doxxing refers to the act of publicly revealing or publishing private or personally identifiable information (such as a home address, phone number, or workplace) about an individual without their consent, typically with the intent to intimidate, harass, or harm them.

sharing and doxxing as significantly more harmful than insults or name-calling. While this study examined general social media platforms (e.g., Facebook or Instagram) rather than social VR specifically, it indicates the need to understand harassment in context and from the victim's perspective. While some work on online games has noted spatial or proximity-based disruptions, such as griefing or avatar crowding, especially in open-world games or social sandbox environments (Boellstorff 2015; Gray 2014), most studies of harassment have emphasized verbal or visual forms (e.g., chat, voice, identity markers) and have not deeply explored the embodied, co-present dynamics that characterize harassment in immersive environments like social VR (Blackwell et al. 2019; Fox and Tang 2017).

While this literature establishes the prevalence and psychological toll of online harassment, an equally important strand of research investigates how platforms and users respond to such harms. These studies highlight punitive moderation strategies as well as emerging frameworks of justice that emphasize repair and accountability. We will review this work next.

2.2 Governance and justice responses to online harassment

In platform governance, conventional responses to harassment have focused on punitive measures such as content removal, user bans, and shadow-banning[3] (Kou 2021; Ma et al. 2023). However, emerging work has called for more expansive justice frameworks that go beyond punishment to include reparative responses like public apologies, financial compensation from the platform, and identity exposure (Schoenebeck et al. 2021; Chen et al. 2024). These approaches reflect broader theories of justice that view harm as not only structural but also emotional, social, and reputational. Drawing from restorative justice (Zehr 2015), procedural fairness (Tyler 2003), and victim-centered design (Schoenebeck et al. 2021), this perspective emphasizes healing, user agency, and meaningful resolution over deterrence alone.

For instance, platforms could offer victims monetary compensation as acknowledgment of their responsibility in hosting the harm and failure to protect, complementing punitive measures with restorative ones (Schoenebeck et al. 2021). Empirical studies show that support for these responses varies across cultural and gender lines. For instance, research focusing on regional differences found that users from non-Western contexts are more likely to favor compensation and identity exposure (Schoenebeck et al. 2023). Separately, studies examining gender have shown that women tend to prefer platform-led interventions like bans and content removal, and express lower support for personally mediated remedies such as payment (Im et al. 2022; Schoenebeck et al. 2023).

Prior research has also documented that some victims of online harassment express a desire for legal consequences for perpetrators, particularly in severe cases, viewing harassment not only as a violation of platform rules but as a criminal act deserving judicial intervention. Qualitative studies have noted that users sometimes turn to law enforcement or courts when platform responses are perceived as inadequate or ineffective (Schoenebeck et al. 2021; Citron 2014). However, these preferences for legal remedies have largely been discussed anecdotally or observed in open-ended responses, rather than systematically measured.

While most studies of online harassment focus on its occurrence and direct psychological effects (Bruckman et al. 1994; Jiang et al. 2021; Stevens et al. 2021), recent work argues for more differentiated models of harm that account for context, repetition, and user interpretation. For example, Schoenebeck et al. (2023) show that perceived harm varies widely not just by type of harassment, but also by factors such as frequency and social framing. Moreover, different forms of perceived harm may not align with preferred remedies (Schoenebeck et al. 2023; Gray 2014). For example, while users may rate certain behaviors like doxxing as extremely harmful, they might simultaneously show lower support for traditional remedies like permanent bans in those specific cases compared to other forms of harassment (Schoenebeck et al. 2023). This suggests that measuring 'impact' as a single construct obscures important variation in how users believe justice should be served for different types of violations. These arguments are echoed in work on justice-centered design, which calls for platforms to recognize emotional and social dimensions of harm beyond punitive measures (Schoenebeck et al. 2021). This is especially relevant in immersive environments, where embodiment and proximity can intensify the affective realism of seemingly minor behaviors (Blackwell et al. 2019; Chen et al. 2024; Weerasinghe et al. 2025).

Despite the richness of this literature, how justice frameworks and user preferences apply in immersive, avatar-based environments is largely unexplored. Social VR introduces unique challenges through real-time embodiment, co-presence, and interpersonal interactions that fundamentally reshape how harassment is enacted and experienced (Freeman et al. 2022). Existing measures (most often punitive tools such as bans) may provide immediate protection but often fail to meet users' broader needs for justice and repair. To address this, our study expands the repertoire of measured responses beyond traditional moderation. This approach responds to calls for governance that recognizes

[3] Shadow-banning is a form of hidden moderation where a user's content is invisibly suppressed from others without their knowledge.

how preferences for remedies vary by context, culture, and type of harm (Schoenebeck et al. 2023, 2021), ultimately seeking to identify more appropriate and just outcomes for targets of harassment in immersive environments.

2.3 Social VR environments and harassment challenges

Social VR platforms support real-time interaction in shared 3D spaces through embodied avatars. These environments offer high social bandwidth via embodiment tracking, spatialized audio, and gestural communication (Maloney et al. 2020; Tanenbaum et al. 2020), enabling immersive interactions that go beyond those afforded by traditional screen-based media (Sykownik et al. 2021; Maloney and Freeman 2020). Users participate in a wide range of activities, from casual hangouts and roleplay to education and emotional support (Piitulainen et al. 2022; Li et al. 2023; Sykownik et al. 2021), and often use avatars to explore identity in expressive, fluid ways (Freeman and Maloney 2021).

Yet the same affordances that make social VR socially rich also amplify its risks. Harassment in these environments is intensified by embodiment and spatial proximity, making unwanted interactions feel more immediate and visceral than in traditional online settings such as forums, multiplayer games, or earlier virtual worlds (Ventura et al. 2021; Freeman et al. 2022), which typically rely on text, voice, or screen-based avatars. While online harassment has long been studied in those contexts (Fox and Tang 2017; Boellstorff 2015), social VR introduces qualitatively distinct forms of harm through real-time co-presence and bodily representation. Research on phantom touch and avatar-based body ownership shows that seemingly virtual actions, such as being touched, crowded, or approached, can feel strikingly real, eliciting strong emotional and physical responses (Ventura et al. 2021; Chen et al. 2024). These effects are consistent with embodiment theory, which suggests that users can come to experience their avatars as extensions of their own bodies, shaping how virtual interactions, such as proximity or touch, are emotionally and somatically registered (Kilteni et al. 2012; Slater and Sanchez-Vives 2016). As a result, avatar-based violations are often experienced as psychologically intrusive, and social VR is increasingly recognized as a space where harassment takes on complex, embodied forms (Chen et al. 2024; Schulenberg et al. 2024; Freeman and Maloney 2021), underscoring the need for empirical research into how users perceive and respond to such embodied harms.

Drawing from qualitative accounts of user experiences, Blackwell et al. (2019) identified three recurring forms of harassment in social VR: verbal harassment, i.e., harassment enacted through spoken language (e.g., slurs, hateful speech); physical harassment, i.e., harassment enacted through avatar-to-avatar embodied actions (e.g., unwanted simulated touching, cornering, or blocking movement); and environmental harassment, harassment enacted by manipulating the shared virtual environment (e.g., displaying explicit or otherwise offensive content on shared screens, placing harassing objects or imagery into a shared space, or altering the space in ways that target a user and make the environment feel hostile). Rather than amplifying existing 2D patterns, immersive harassment is increasingly conceptualized as a distinct phenomenon shaped by presence, affective realism, and the psychological salience of avatar embodiment (Blackwell et al. 2019; Chen et al. 2024; Freeman and Maloney 2021).

Recent studies confirm the prevalence and impact of such behaviors, with up to 95% of users reporting having witnessed or experienced harassment in social VR (Weerasinghe et al. 2025). Marginalized groups, including women, LGBTQ+ individuals, and disabled users, are particularly at risk (Schulenberg et al. 2024; Freeman et al. 2022). Safety features such as muting, blocking, and personal space boundaries are commonly used, yet often perceived as inadequate, inconsistently applied, or overly reactive (Weerasinghe et al. 2025). Prior research has primarily focused on how users adopt and engage with these tools (Blackwell et al. 2019; Weerasinghe et al. 2025), with comparatively less attention to how users interpret the severity of different harassment types or evaluate potential remedies beyond immediate mitigation. Moreover, despite increasing documentation of embodied harassment in social VR, there remains a lack of normative or legal or policy frameworks to guide how such incidents should be addressed (Richardson et al. 2024), raising important questions about accountability in immersive environments.

Building on prior qualitative accounts that identified verbal, physical, and environmental harassment in social VR (Blackwell et al. 2019), our study examines these types across three dimensions: frequency, harmfulness, and governance response preferences. This focus moves beyond prevalence reporting to show how different forms of harassment are experienced and evaluated, and how gender shapes these perceptions. In doing so, we extend online harassment research to immersive contexts, where embodiment and co-presence reshape both the experience of harassment and expectations for justice, and provide empirical grounding for context-sensitive, user-centered safety interventions.

3 Methods

To investigate user perceptions and experiences of harassment in social VR, we designed and conducted an anonymous online survey targeting adult (18+) users of social VR platforms. Our goal was to understand how users of social VR platforms experience and evaluate different forms of harassment, and how they perceive various possible responses to such incidents. The study is considered an exemption by our institution's ethics review board. This study was not preregistered. This study adopts an exploratory design. Although prior work on harassment in 2D online environments could inform directional expectations, the unique affordances of social VR, such as avatar embodiment, identity fluidity, and spatial co-presence, make it uncertain whether established patterns transfer to immersive contexts. We therefore did not formulate confirmatory hypotheses a priori, but aimed to establish descriptive baselines. We compare our findings with the 2D literature in the Discussion to inform future confirmatory research. To strengthen transparency and reproducibility, we will release the fully anonymized dataset as complementary material upon acceptance.

3.1 Participant recruitment and demographics

The survey was administered via Prolific.[4] Participants were compensated at a rate of 9.52 UK pounds/hour, consistent with Prolific's fair pay guidelines. A total of 297 participants initially completed the survey. However, we removed incomplete responses and data that did not meet our eligibility criteria, such as participants under 18 or those without prior social VR experience. After applying these filters, we obtained a final dataset of 215 valid responses. In terms of age distribution, 53 participants (24.65%) were between 18 and 24 years old, 107 (49.77%) were between 25 and 34 years old, 29 (13.49%) were between 35 and 44 years old, 18 (8.37%) were between 45 and 54 years old, 5 (2.33%) were between 55 and 64 years old, and 3 (1.40%) were 65 years or older. Regarding gender distribution, 124 participants (57.94%) identified as men, 87 (40.65%) identified as women, 3 (1.40%) identified as non-binary, and 1 participant preferred not to disclose their gender. In terms of regional distribution, 117 participants (54.42%) were from Europe, 31 (14.42%) were from North America, 8 (3.72%) were from South America, 52 (24.19%) were from Africa, 3 (1.40%) were from Oceania, and 4 (1.86%) were from Asia.

[4] Prolific, https://www.prolific.com/, is a well-regarded online data collection platform known for its ability to recruit anonymous participants and deliver high-quality, reliable data.

3.2 Survey measures

Participants were provided with a brief description of social VR, following the work of Schulenberg et al. (2024), which highlighted key features such as avatars, HMD, and popular platforms like VRChat and Rec Room. Participants were asked two screening questions: (1) whether they had previously used social VR with a VR headset, and (2) whether they were at least 18 years old. Only those who answered "Yes" to both questions were allowed to proceed with the survey. Those who failed to meet either criterion were automatically excluded. Prior social VR exposure was then measured using a self-reported usage frequency item (i.e., Not at all, Once a month or less, A few times a month, A few times a week, Daily). Participants who selected "Not at all" were excluded, serving as an additional check to ensure prior social VR experience.

The measures of harassment followed prior classifications of harassment in social VR (Blackwell et al. 2019), distinguishing three main types: verbal, physical, and environmental. These were assessed along three key axes: frequency of experience, user perception, and response preferences. To minimize potential order effects, the order of harassment types and remedies was randomized across participants. Clear definitions and examples of each type of harassment were included within the survey to reduce ambiguity. For each type of harassment, participants reported how often they had experienced it using a five-point ordinal scale (i.e., Never, Once, A few times, Several times, Many times). Harmfulness was measured on a five-point scale (i.e., Not harmful at all, Slightly harmful, Moderately harmful, Quite harmful, Extremely harmful).

Participants rated the desirability of five possible responses to each harassment type on a 5-point Likert scale ("Not at all desirable" to "Extremely desirable"). These responses were informed by prior work showing that users' preferences extend beyond punishment to include fairness, accountability, and repair (Schoenebeck et al. 2023, 2021; Im et al. 2022). From the broader set identified in earlier studies, we further refined and tailored them to the dynamics of immersive, avatar-based environments. These responses included: *Banning the perpetrator's account*, representing the prevailing punitive model of platform moderation. *Paying compensation to the victim*, a form of economic justice/material redress addressing losses experienced by targets. *Requiring public apology from the perpetrator*, a restorative-justice remedy emphasizing acknowledgment of harm and emotional repair. *Revealing the perpetrator's real name and photograph*, an identity/accountability sanction for deterrence via exposure (shaming-type measure). *Taking legal action by law enforcement*, representing a legal intervention that recognizes harassment to be treated as

criminal offenses. We added this option to extend the repertoire beyond platform-based governance, given the increasing recognition that immersive harassment can cross into legally actionable territory.

Demographic information was collected on age, gender identity, country of residence, and racial/ethnic identification to allow exploration of subgroup differences. Responses were not collected separately for specific platforms. We interpret the results as reflecting participants' general experiences in social VR.

3.3 Data analysis

All analyses were conducted using non-parametric statistical methods, as the key outcome variables were measured on Likert-type ordinal scales and assumptions of normality could not be guaranteed. Descriptive statistics were first computed to summarize distributions of reported harassment experiences, perceived harm, and the desirability of response strategies. To compare differences across harassment types (verbal, physical, and environmental) and across response strategies (ban, apology, legal action, pay money, reveal ID), we employed Friedman tests as a non-parametric alternative to repeated-measures ANOVA. Where omnibus tests indicated significant effects, we conducted post-hoc Wilcoxon signed-rank tests with Bonferroni correction to identify specific pairwise differences.

Gender differences were examined using Mann–Whitney U tests, which provide a robust non-parametric alternative to independent-samples t-tests. These analyses were applied separately to harassment frequency, perceived harm, and desirability ratings of response strategies. To further explore subgroup patterns, we also conducted within-gender analyses by running Friedman tests separately for male and female participants, followed by Wilcoxon signed-rank tests where appropriate.

Effect sizes were reported to facilitate interpretation of the magnitude of findings. For Wilcoxon signed-rank and Mann–Whitney U tests we reported *r*, calculated as $Z/\sqrt{N}$. Following conventional benchmarks, values around .10 were interpreted as small, .30 as medium, and .50 or above as large effects. All analyses were conducted in R, and results were normally visualized through figures and tables to illustrate key patterns in harassment frequency, perceived harm, and governance preferences. As a robustness check, we also conducted a full set of parametric analyses (e.g., repeated-measures ANOVA, independent-samples *t*-tests). These yielded a similar overall pattern of results and are reported in the Supplementary Materials (Sections S1–S3).

3.4 Positionality statement

We include a positionality statement to acknowledge how our identities, experiences, and investments may shape the framing and interpretation of this work. All authors have prior experience in engaging with social VR communities, either through personal participation or previous research endeavors. Among them, three members have specific expertise in studying toxicity and moderation practices in immersive environments, which informs the framing of this work. Our team includes both male and female researchers, which provides a balanced perspective particularly relevant to this study's focus on gendered differences in harassment perception and governance preferences. While we recognize that our own identities and experiences inevitably shape our interpretations, we have aimed to foreground participant voices and empirical evidence in our analysis. We view this project as part of an ongoing commitment to producing research that is context-sensitive and attentive to the diverse realities of social VR users.

4 Results

This section presents findings on the frequency, harmfulness, and preferred remedies for harassment in social VR. Additional descriptive statistics and breakdowns are provided in the Appendix, including Appendix 1 for frequency; Appendix 2 for harmfulness; Appendix 3 for perceived desirability of response strategies.

4.1 Frequency of harassment

4.1.1 Comparison across harassment types

Verbal harassment was the most frequently experienced type: only 27.0% reported never experiencing it, whereas 20.0% reported frequent occurrences (i.e., "Several times" or "Many times"). In contrast, physical harassment was least common (63.3% reported never experiencing it; 7.9% reported frequent occurrences), with environmental harassment falling in between (Fig. 1). Consistent with these distributions, mean frequency ratings (on a scale from 1 = Never to 5 = Many times) showed the same ordering (verbal: $M = 2.52$, $SD = 1.21$, i.e., between "Once" and "A few times"; environmental: $M = 2.10$, $SD = 1.15$, i.e., approximately "Once"; physical: $M = 1.72$, $SD = 1.10$, i.e., between "Never" and "Once").

A Friedman test confirmed that harassment frequency differed significantly across the three types, $\chi^2(2) = 101.00$, $p < .001$ (Kendall's $W = .24$). Post-hoc Wilcoxon signed-rank tests (with Bonferroni correction) showed all pairwise

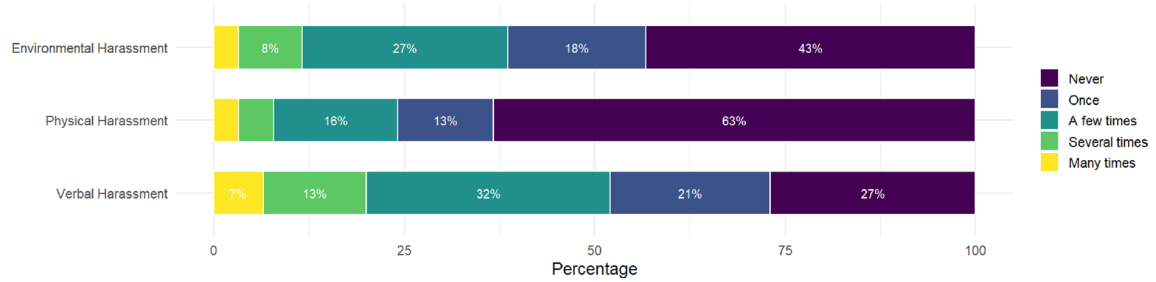


Fig. 1 Distribution of harassment frequency by type

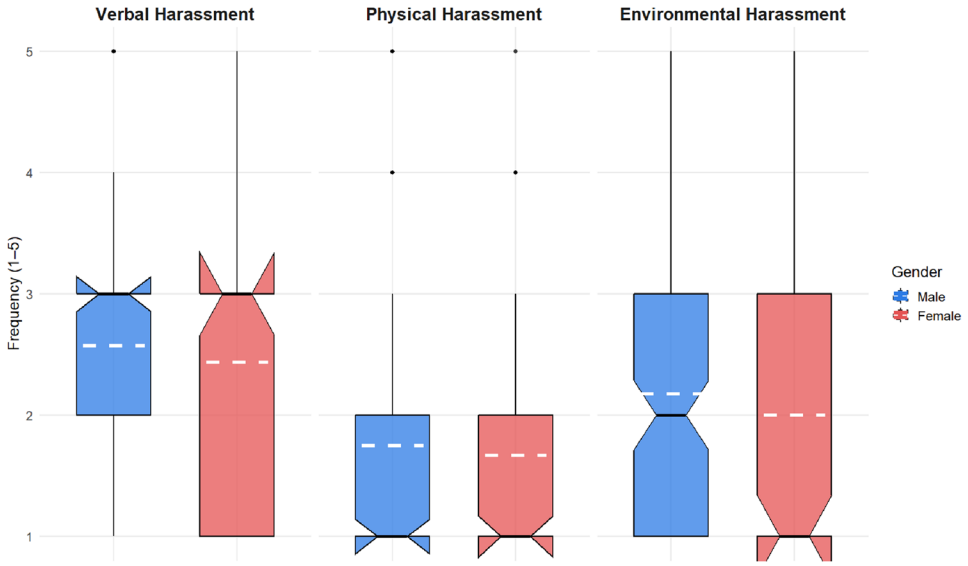


Fig. 2 Harassment frequency ratings by gender across harassment types

differences were significant: verbal harassment versus physical harassment ($p < .001$, $r = .60$); verbal harassment versus environmental harassment ($p < .001$, $r = .40$); environmental harassment versus physical harassment ($p < .001$, $r = .38$).

4.1.2 Gender as a factor

4.1.3 Gender differences in frequency

Figure 2 shows harassment frequency ratings by gender across harassment types. Overall, male and female participants reported similar frequencies of verbal, physical, and environmental harassment, with largely overlapping distributions across all three contexts.

We further examined whether the frequency of harassment differed statistically by gender using Mann–Whitney U tests for each harassment type. After applying a Bonferroni correction ($\alpha = .05/3 = .017$), no statistically significant gender differences were observed across verbal, physical, or environmental harassment (see Appendix 1, Table 4). These findings suggest that male and female participants reported similar frequencies of harassment across all three types.

4.1.4 Within-gender comparisons of harassment frequency

Nonparametric Friedman tests revealed significant differences in harassment frequency across types within both male and female participants (male: $\chi^2(2) = 56.10$, $p < .001$; female: $\chi^2(2) = 42.90$, $p < .001$). Post-hoc Wilcoxon signed-rank tests with Bonferroni correction indicated that, in both gender groups, verbal harassment was reported more frequently than physical and environmental harassment, and environmental harassment was reported more frequently than physical harassment. Omnibus Friedman tests and Bonferroni-corrected post-hoc comparisons are reported in Table 1.

4.1.5 Exposure-adjusted robustness check (ordinal mixed-effects model)

Because participants differed in prior social-VR usage frequency, which may proxy exposure opportunity, we conducted a robustness check using a cumulative link mixed

Table 1 Post-hoc Wilcoxon signed-rank tests of harassment frequency within gender groups

Gender	Type	Comparison	p-value	r
Male	Verbal	Verbal versus Physical	$< .001$***	.60
	Physical	Verbal versus Environmental	$< .001$***	.38
	Environmental	Environmental versus Physical	$< .001$***	.40
Female	Verbal	Verbal versus Physical	$< .001$***	.60
	Physical	Verbal versus Environmental	$< .001$***	.43
	Environmental	Environmental versus Physical	.003**	.36

*$p < .05$, **$p < .01$, ***$p < .001$. Adjusted p-values reported

model (logit link) predicting 5-point harassment-frequency ratings across the three harassment types. The model included harassment type and usage frequency as fixed effects, and a participant-level random intercept to account for repeated measures. Results mirrored the non-parametric comparisons reported above: relative to verbal harassment, physical harassment showed substantially lower odds of higher frequency ratings (OR = 0.08, 95% CI [0.05, 0.13]), and environmental harassment was also lower (OR = 0.30, 95% CI [0.19, 0.45]). Usage frequency was a significant predictor: compared to participants using social VR once a month or less, using a few times a week increased the odds of reporting a higher harassment-frequency category (OR = 4.17, 95% CI [1.32, 13.11]), and daily use increased these odds further (OR = 16.28, 95% CI [3.37, 78.66]). These exposure-adjusted effects are visualized in Fig. 3.

4.2 Perceived harm of harassment types

4.2.1 Comparison of harassment types

As shown in Fig. 4, verbal harassment was most commonly perceived as moderately or extremely harmful (29.3% and 24.7%, respectively), with only 7.9% rating it as not harmful at all. Physical harassment exhibited a more polarized pattern: 30.7% rated it as extremely harmful, while 19.1% rated it as not harmful at all. Perceptions of environmental harassment fell between these two patterns, showing a more even distribution across harm categories (Fig. 4). Consistent with these distributions, mean harm ratings (on a scale from 1 = Not harmful at all to 5 = Extremely harmful) were highest for verbal harassment ($M = 3.46$, $SD = 1.25$, i.e., between "Moderately harmful" and "Quite harmful"), followed by physical harassment ($M = 3.26$, $SD = 1.51$, i.e., approximately "Moderately harmful") and environmental harassment ($M = 3.16$, $SD = 1.36$, i.e., approximately "Moderately harmful").

A Friedman test indicated a significant overall difference in perceived harm across harassment types, $\chi^2(2) = 9.16$, $p = .010$. Post-hoc Wilcoxon signed-rank tests with Bonferroni correction showed that verbal harassment was rated as more harmful than environmental harassment ($p < .001$, $r = .26$), whereas the differences between verbal and physical harassment ($p = .071$, $r = .15$) and between physical

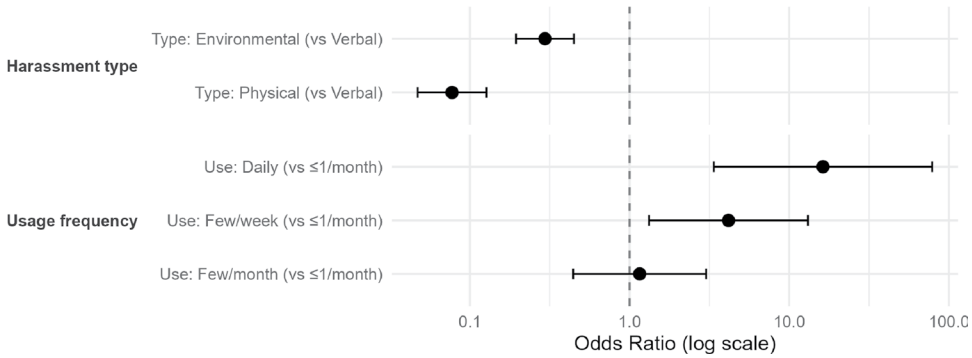


Fig. 3 Forest plot of odds ratios from the cumulative link mixed model predicting harassment frequency, adjusted for usage frequency

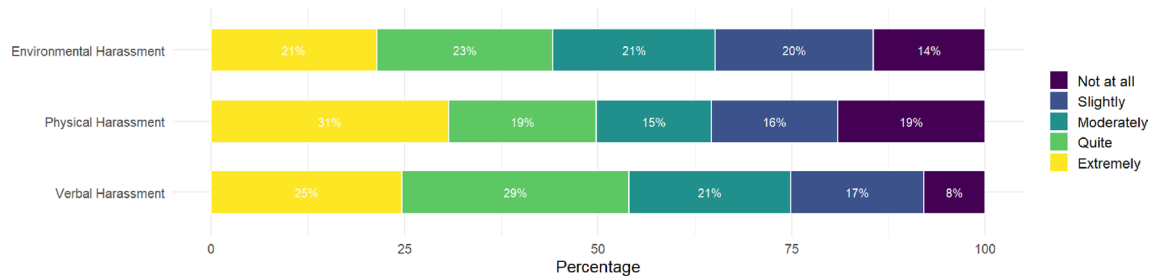


Fig. 4 Perceived harm levels for verbal, physical, and environmental harassment



Fig. 5 Perceived harm of verbal, physical, and environmental harassment by gender

Table 2 Post-hoc comparisons of perceived harm ratings within gender groups

Gender	Comparison	*p*-value	*r*
Male	Verbal versus Physical	$p = .016$*	.25
	Verbal versus Environmental	$p = .004$**	.31
	Physical versus Environmental	$p = 1.000$ ns	.03
Female	Verbal versus Physical	$p = 1.000$ ns	.01
	Verbal versus Environmental	$p = .163$ ns	.21
	Physical versus Environmental	$p = .046$*	.26

*$p < .05$, **$p < .01$, ***$p < .001$. Adjusted *p*-values reported; ns = non-significant

Table 3 Spearman correlations between harassment frequency and perceived harm

Harassment type	*p*-value	r_s
Verbal	0.148	0.142
Physical	1.000	−0.037
Environmental	1.000	0.035
Overall	1.000	0.014

r_s denotes Spearman's rank correlation. Reported *p*-values are Bonferroni-corrected

and environmental harassment ($p = .303$, $r = .11$) were not statistically significant.

4.2.2 Effect of gender on the perceived harm

Figure 5 shows perceived harm ratings by gender across harassment types. Overall, female participants rated all three harassment types as more harmful than male participants. This pattern was consistent across verbal, physical, and environmental harassment.

To assess gender differences, we conducted three Mann–Whitney *U* tests (Bonferroni-adjusted $\alpha = .017$). After correction, perceived harm was significantly higher among female participants for all three harassment types, with effect sizes in the small-to-medium range (see Appendix 2, Table 5 for full test statistics). The gender gap was most pronounced for physical harassment, while verbal and environmental harassment showed comparably smaller but reliable differences.

4.2.3 Perceived harm within gender groups

Separate Friedman tests were conducted for each gender group. For males, results indicated a significant difference in harmfulness across harassment types, $\chi^2(2) = 7.19$, $p = .028$. Post-hoc Wilcoxon signed-rank tests with Bonferroni correction showed that verbal harassment was rated as more harmful than both physical and environmental harassment, while physical and environmental harassment did not differ significantly (see Table 2). For females, the Friedman test was not significant, $\chi^2(2) = 4.91$, $p = .086$, but exploratory post-hoc comparisons suggested that physical harassment was perceived as more harmful than environmental harassment; comparisons involving verbal harassment were nonsignificant.

4.2.4 Relationship between harassment frequency and perceived harm

We examined whether perceived harm covaries with how frequently participants experienced harassment. Because both variables were measured on 5-point ordinal scales, we computed Spearman rank correlations between frequency and harm by harassment type (Table 3). After Bonferroni correction, we found no evidence of an association between harassment frequency and perceived harm: correlations were small and non-significant across all harassment types. Verbal harassment showed the largest (but still weak) positive trend ($\rho = 0.14$, $p = .148$), whereas correlations for physical and environmental harassment were near zero.

4.3 Desirability of responses

4.3.1 Desirability across harassment types

Figure 6 shows a heatmap of the mean desirability scores for different response strategies across three types of harassment. Banning is the most desirable response across all harassment types (on a scale from 1 = Not at all desirable to 5 = Extremely desirable), with the highest scores (M=4.09, i.e., between "Very desirable" and "Extremely desirable"), followed by Legal Action (M=3.37, i.e., between "Moderately" and "Very desirable") and Apology (M=2.99, i.e., approximately "Moderately desirable"), whereas Pay Money (M=2.57) and Reveal ID (2.67) receive the lowest ratings, both falling between "Slightly" and "Moderately desirable".

A Friedman test confirmed that desirability differed significantly across strategies, $\chi^2(4) = 228.00$, $p < .001$. Post-hoc comparisons (Bonferroni-corrected) showed that Ban was rated significantly higher than the remaining strategies, and Legal Action was rated significantly higher than Apology, Pay Money, and Reveal ID. Apology was preferred over both Pay Money and Reveal ID. Full pairwise post-hoc results and effect sizes are reported in Table 6.

We conducted separate Friedman tests for each harassment type to evaluate differences in the perceived desirability of five response strategies. All three tests yielded statistically significant results, indicating that participants significantly differentiated between strategies: Verbal Harassment: $\chi^2(4) = 239.00$, $p < .001$; Physical Harassment: $\chi^2(4) = 288.00$, $p < .001$; Environmental Harassment: $\chi^2(4) = 211.00$, $p < .001$. Post-hoc Wilcoxon signed-rank tests with Bonferroni correction showed that *Ban* was consistently rated as the most desirable strategy across all harassment types, significantly outperforming all others ($p < .001$ in all cases). *Legal Action* was a strong secondary option, but in the case of environmental harassment, the difference between *Legal Action* and *Apology* was not statistically significant. Meanwhile, *Pay Money* and *Reveal ID* remained the least preferred, with no significant difference found between them in any context ($p > .85$ across all types). Full test statistics (including U, adjusted p-values, and r) are reported in Appendix 3 (Table 7).

4.3.2 Gender as a factor

4.3.3 Gender differences in desirability ratings

Figure 7 provides an overview of gender differences in desirability ratings across harassment types and response strategies. Across all harassment types, female participants generally rated most strategies as more desirable than male participants, with the largest differences observed for Legal Action and Reveal ID. These differences were most pronounced for physical and environmental harassment. In contrast, Pay Money showed no gender differences across any harassment type. A distributional view of these ratings is provided in Appendix 4 (Fig. 13).

We first conducted Mann–Whitney U tests comparing male and female participants' ratings aggregated across harassment types (all types combined). This provides a high-level summary of whether gender is associated with desirability ratings independent of harassment type. Overall, female participants rated all response strategies as more desirable than male participants, with statistically significant differences for all strategies except Pay Money ($p = .341$, $r = .04$). The largest gender difference was found for Legal Action, followed by Reveal ID, Apology, and Ban. Full results are shown in Table 8.

To assess whether these overall gender differences were consistent across contexts (i.e., not driven by a single harassment type), we then repeated the Mann–Whitney U tests within each harassment type (verbal, physical, and environmental). The same pattern emerged across types: female participants rated *Ban*, *Apology*, *Legal Action*, and

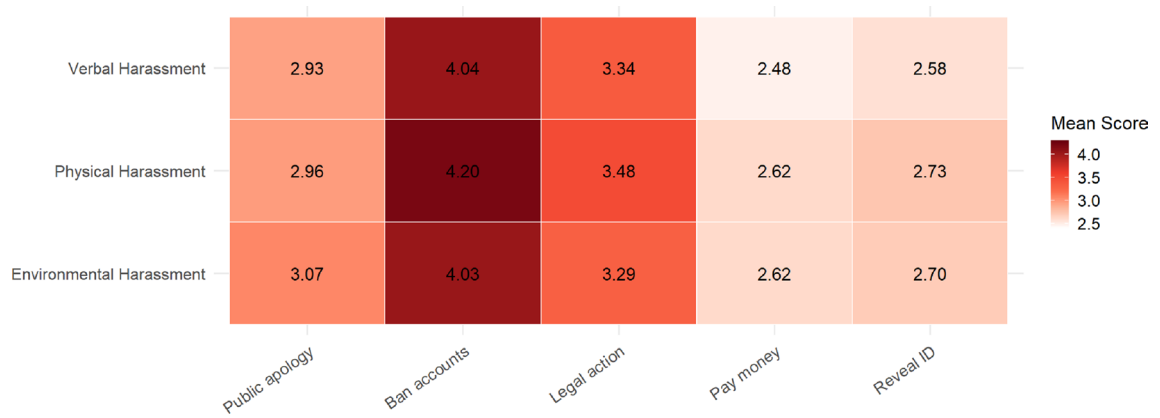


Fig. 6 Heatmap of mean desirability ratings for response strategies (Apology, Ban, Legal Action, Pay Money, Reveal ID) across harassment types

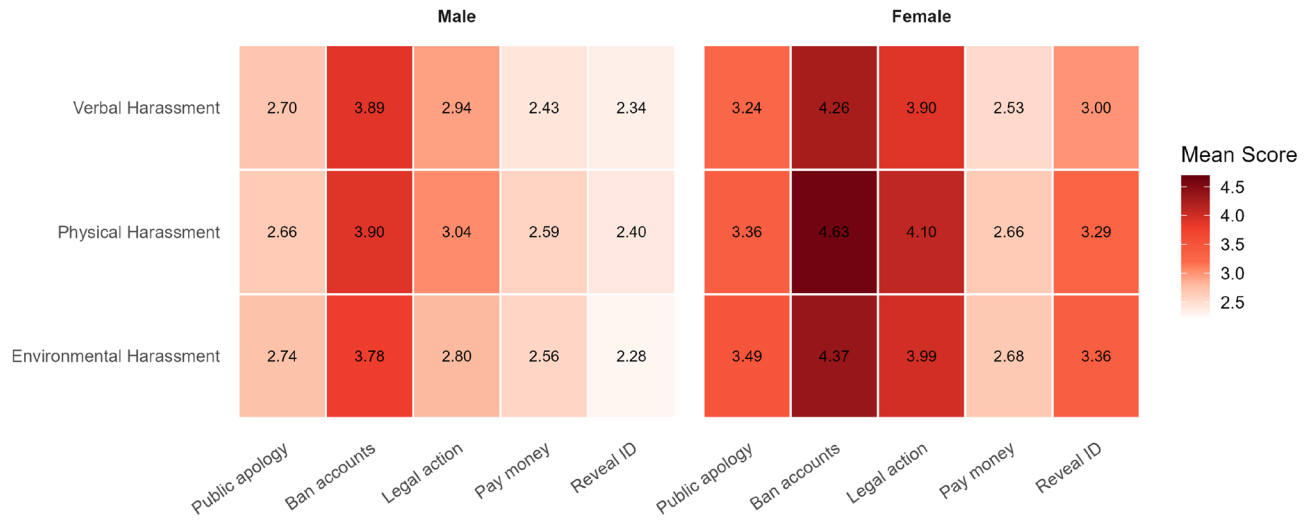


Fig. 7 Gender differences in desirability ratings across harassment types and response strategies. Cells show mean desirability ratings; darker shading indicates higher desirability

Reveal ID as more desirable than male participants, whereas *Pay Money* showed no significant gender difference in any type. The most significant difference in desirability is the legal action taken against physical harassment, with the biggest effect sizes.Full test statistics (including U, Bonferroni-adjusted p-values, and effect sizes r) are reported in Appendix 4 (Table 9).

4.3.4 Within-gender comparisons of response-strategy desirability

To characterize within-gender preference structures, we further examined within-gender differences in the perceived desirability of response strategies for each harassment type. Results of the Friedman tests were significant for both males and females across verbal, physical, and environmental harassment, indicating systematic differences in strategy preferences, see Table 10 (Appendix 4) for the details.

While the overall ordering was similar for both genders, female participants showed stronger differentiation among the remaining strategies beyond banning. In particular, *Legal action* and *Apology* tended to receive comparatively higher endorsement among women than among men across contexts. Complete Bonferroni-adjusted post-hoc Wilcoxon signed-rank results and effect sizes are reported in Appendix 4 (Table 11).

5 Discussion

5.1 Harassment in social VR in context

5.1.1 Prevalence and variation

Verbal harassment emerged as the most frequently reported type in this study. This aligns with broader research on online harassment, which consistently identifies verbal abuse as the dominant form across platforms, including social media, multiplayer online games, and virtual communities. It indicates that despite the novel interaction modes VR offers, traditional forms of abuse (e.g. slurs, insults, voice chat abuse) remains a dominant threat. In contrast, physical harassment, a form of harm unique to immersive environments, was reported less commonly. However, this lower relative frequency should not be misinterpreted as a lack of significance. Despite occurring less often than verbal abuse, physical harassment remains a pervasive issue, with 37% of our sample reporting at least one such experience (see Fig. 1). Furthermore, the fact that it occurs less often than verbal abuse should not diminish the gravity of these encounters, as even less frequent physical violations can have a profound impact in immersive environments. Our study demonstrates that there is considerable variability in users' perceptions regarding physical harassment. Unlike verbal harassment, which was broadly viewed as harmful, physical harassment received ratings concentrated at both extremes: "not harmful at all" and "extremely harmful." This polarized perception can be explained by differences in how users interpret physical interactions in virtual reality. Some users may perceive physical interactions within VR as inherently unreal or lacking tangible consequences compared to

real-world physical assault, thus rating physical harassment as minimally harmful. Conversely, other users who strongly experience self-avatar embodiment may experience intense psychological distress from virtual physical harassment. Users with a high sense of embodiment might subconsciously interpret such virtual violations as if they were happening to their real bodies, which heightens the psychological impact. This phenomenon can be further intensified by experiences such as phantom tactile sensations (the illusion of feeling touch without actual physical contact (Chen et al. 2024)), causing some individuals to perceive physical harassment as particularly harmful despite the absence of real physical contact. Beyond the psychological effects of embodiment, this polarization may also reflect the safety concerns users bring from their offline lives into the headset (Blackwell et al. 2019). Research indicates that virtual experiences are often interpreted through the lens of offline social positioning and prior life experiences (Maloney and Freeman 2020). For instance, individuals who navigate the physical world with a heightened fear of sexual violence or bodily violation, a highly gendered experience, may perceive unwanted virtual proximity or simulated touch as a significant threat to their personal safety. In this context, the virtual violation is not viewed in isolation but as an extension of power dynamics and threats to bodily autonomy from the offline world. As a result, the high harm ratings in our study likely represent users who feel that their virtual safety and physical safety are deeply connected, whereas lower ratings may come from those who do not associate such interactions with offline physical risks.

5.1.2 Gendered dynamics

While prior research consistently indicates that female users generally experience higher rates of harassment in online gaming and social media (Bartlett et al. 2014; Jane 2014), our findings suggest no significant differences in self-reported frequency of verbal, physical, or environmental harassment between men and women in social VR. This result implies that, within the context of social VR, harassment is experienced at comparable rates across genders.

This divergence from other settings may come from unique characteristics of social VR environments. A key contributor to this may be the prevalent practice of avatar gender-switching. Many users adopt different avatars for purposes such as curiosity, identity exploration, self-expression, or entertainment (Freeman and Maloney 2021; Freeman and Acena 2022). Consequently, an avatar's presented gender often does not reflect the user's actual gender. It potentially masks traditional patterns of gender-based harassment. Additionally, the widespread voice-changing and mute practice in social VR may complicate traditional harassment dynamics (Chen et al. 2025; Freeman and Maloney 2021).

Despite no significant gender differences in harassment frequency, female participants consistently rated all types of harassment as significantly more harmful than male participants. These results align closely with existing literature that women generally report higher psychological distress from online harassment compared to men (Vogels 2021; Im et al. 2022). Notably, the largest gender difference was observed in perceptions of physical harassment. Male participants reported significantly lower perceived harm than female participants, with this condition showing the highest r value among the three types of harassment. While female participants rated both verbal ($M = 3.75$) and physical harassment ($M = 3.72$) similarly (both perceived as highly harmful), male participants demonstrated a larger discrepancy, viewing physical harassment ($M = 2.92$) as notably less harmful than verbal harassment ($M = 3.24$). This discrepancy likely comes from both lived experiences and broader socio-cultural factors. Women tend to disproportionately experience physical harassment or sexual assault in everyday life. It may make them more attuned to boundary violations, even in virtual environments (Schulenberg et al. 2023). In addition, due to prolonged socio-cultural conditioning (such as through socialization practices, media portrayals, and even educational messaging that emphasize women's need for vigilance), unwanted virtual contact can feel closely tied to real-world threats, amplifying its perceived harm. Conversely, men's much higher perceived harm ratings for verbal harassment, compared to physical harassment, align with evidence showing males tend to feel more threatened by verbal attacks targeting their identity, competence, status, or intelligence (Bekiari and Hassanagas 2015; Stanaland and Gaither 2020). By contrast, physical harassment in VR for them is more likely to be dismissed as playful or unrealistic, and therefore less damaging.

This study offers several contributions to the growing literature on online and virtual harassment. First, it provides empirical evidence on the prevalence and perception of harassment types specifically within social VR environments. Second, our findings on harassment perception polarity, especially around physical harassment, highlight the unique psychological mechanisms at play in VR. Because physical harassment is a dimension largely absent from 2D online environments, its impact cannot be directly inferred from existing 2D research. While the demographic patterns of harm (e.g., gender differences) may align with traditional findings, the nature of the violation is distinct; it involves an intrusion into personal space and a threat to bodily autonomy that is unique to immersive, embodied media. Third, by demonstrating gender-based differences in harm perception rather than harassment frequency, we suggest that social

VR platforms may obscure traditional gender disparities in exposure but not necessarily in impact. This distinction contributes to discussions about avatar-based identity masking and how it may complicate conventional assumptions about gendered experiences online (Freeman and Maloney 2021; Freeman and Acena 2022).

5.2 Rethinking governance frameworks for immersive harassment

The strong user support for legal action in response to severe VR harassment feeds into an ongoing debate: Should virtual misconduct in VR (or metaverse) be addressed purely as a platform governance issue, or are there circumstances where it becomes a law enforcement matter? Traditionally, harassment in online platforms has been managed through community guidelines and moderation, such as content removal, user suspensions, and bans, with the view that platforms govern user behavior in their spaces.

Our findings that users endorse legal action for physical harassment in VR contribute empirical weight to these discussions. They suggest that a segment of users view certain virtual transgressions as fundamentally unacceptable to the point of meriting criminal or civil consequences, not just a ban. While there have been investigations (McGlynn and Rigotti 2025), this does not yet amount to a widely settled or uniformly actionable legal framework for avatar-mediated misconduct. However, if virtual groping becomes more common and public pressure grows, we may similarly see legal systems evolve to recognize virtual sexual assault as a punishable offense widely. That said, significant challenges remain. Jurisdictional uncertainty (who has authority over a transnational virtual incident) and evidentiary issues (how to prove a virtual interaction in court) are unresolved hurdles. Moreover, not all users may truly want to entangle the police or courts with their online experiences. Supporting for "legal action" in our survey might reflect a desire for seriousness and accountability, rather than an actual appetite for courtroom proceedings. Nonetheless, even in the absence of clear legal pathways, the endorsement of legal involvement may signal a boundary-making move: rejecting the idea that harms occurring in the VR should remain confined to platform governance alone.

These findings carry implications for how we design governance frameworks in immersive platforms. First, platform companies may need to recognize that user expectations around harassment governance are evolving. It is no longer sufficient to treat harassment as a minor community management issue; in some cases, users expect platforms to facilitate or at least not obstruct external justice. For platform policy, this could mean developing clear protocols for when and how to escalate certain incidents (e.g., sexual assault in VR, credible violent threats) to law enforcement, in line with user sentiments. At the very least, platforms should empower users with tools to document and report severe abuse so that if a victim wishes to pursue legal action, they can do so with evidence. This is analogous to how some social networks now cooperate with authorities on child exploitation cases (Quayle 2020). Our study suggests a similar boundary may need to be drawn for VR harassment. Platforms might also consider incorporating warnings and education about the possibility of legal consequences for extreme misconduct, reinforcing that virtual actions can have real-life repercussions.

At the same time, the continued primacy of banning and moderation in user preferences reminds us that robust internal governance is still the frontline defense. Social VR platforms should ensure their harassment reporting and banning systems are effective, transparent, and rapid, as users rely on these for immediate protection. As Schoenebeck et al. (2023) argues, a one-size-fits-all approach to moderation is inadequate. Platforms should tailor their policies to the specific types of harm and the affected communities. For example, a lewd comment and a simulated sexual assault in VR are both harassment, but vastly different in impact; the former might be handled with a warning or short ban, whereas the latter might justify a permanent ban and informing the user of external options. Governance frameworks would benefit from being victim-centered, taking into account the trauma and harm of different harassment incidents. This could involve integrating trauma-informed practices or restorative justice mechanisms alongside punitive measures. Although participants expressed limited support for remedies such as apologies or financial compensation, it is possible that offering these in combination (e.g., an apology alongside a ban) could provide victims with a greater sense of closure, provided such processes are implemented with care, as poorly executed apologies may have adverse effects. Ultimately, a holistic governance framework might offer a menu of responses: immediate safety tools (e.g., mute, block), platform enforcement (e.g., ban, offender flagging), support options for victims (e.g., peer support communities), restorative options (facilitated apologies or mediation if the victim desires), and channels for legal escalation when appropriate. The key is that the framework should be flexible and responsive to the context of the harm, as theory and research increasingly emphasize. In a broader perspective, these developments in social VR governance tie into ongoing re-examination of platform power and responsibility. For years, platforms operated under a paradigm of maximum autonomy. They set their own rules and are largely shielded from liability (e.g., via Section 230 in the US) (Filatova-Bilous 2021). However, as online spaces become more immersive and the boundary between "virtual" and

"real" harm further dissolves, there is growing pressure on platforms to take on duty-of-care responsibilities akin to real-world institutions.

5.3 Limitations and future works

This study offers insights into harassment experiences in social VR, yet several limitations should be acknowledged. Our findings are based on self-reported survey responses, which are subject to memory bias, social desirability bias. Additionally, while we included gender as a key variable, other intersecting identities such as race, sexual orientation, or disability status, were not deeply analyzed, which may overlook dimensions of harassment experiences. We did not collect platform-specific reports. The choice follows previous research on harassment in social VR, treating social VR as a cross-application ecology (Blackwell et al. 2019). While this approach has the advantage of focusing on shared affordances rather than single-platform effects, it limits our ability to examine platform-level differences. Our study used a single-item measure to assess different constructs. While it is common in exploratory work and aligns with prior research on online harassment and harm in general (Im et al. 2022; Strong et al. 2019), it may limit the depth of the data. Consequently, our results provide a broad overview of harm but may lack the nuance required to identify which specific types of impact were most significant for different users. Additionally, the harmfulness item asked participants to rate harassment "in general, based on your thoughts or experiences," which does not strictly separate personal harm from a general judgment about the behavior's severity. Participants may have drawn on either or both perspectives when responding. While this blended framing is consistent with how perceived harm has been operationalized in prior work (Schoenebeck et al. 2023), future research should consider disentangling personal and general evaluations of harm. Third, although participants were drawn from multiple geographic regions, the sample was unevenly distributed across regions, with relatively small numbers from several areas. The study, therefore, does not support region-level comparisons, and conducting inferential analyses across regions would risk overinterpretation. Lastly, we did not collect data on participants' prior experiences with or attitudes toward harassment in the offline world. Since the perceived harm of virtual physical violations may be influenced by a user's offline history and their general sense of safety, future research should explore how offline vulnerability and prior experiences of harassment shape the psychological impact of violations in Social VR.

Our findings indicate that perceived harm differs by gender. Future research should explore how individual differences in identity salience, prior trauma, or cultural norms shape users' sensitivity to embodied harassment. For example, experimental studies that manipulate avatar appearance and control schemes may offer deeper insights into embodied vulnerability in VR spaces. Additionally, the unique forms and effects of harassment in VR environments challenge traditional frameworks derived from text-based or 2D platforms. Future theoretical work should develop new conceptual models that incorporate presence, embodiment, and spatial dynamics, potentially drawing from embodiment theory and digital phenomenology. Future work with more balanced regional sampling and platform-specific analysis is needed to systematically examine how cultural and regional contexts, as well as platform specifics, shape perceptions of harassment in social VR.

6 Conclusion

This study examined harassment in social VR, comparing verbal, physical, and environmental forms in terms of frequency, perceived harm, and preferred remedies. Verbal harassment was the most common and broadly harmful, while physical harassment, unique to embodied co-presence, was less frequent but polarized in its impact. Gendered differences emerged not in frequency but in perception: women rated all forms of harassment as more harmful and showed stronger support for punitive and legal responses. Across users, bans were the most desirable remedy, with legal action also receiving notable support, while compensation and identity exposure were the least favored. As immersive technologies deepen co-presence and blur the boundary between virtual and material experience, governance models must evolve in tandem. These findings suggest that user judgments of harm and preferred remedies can guide the development of safety frameworks in social VR that acknowledge embodiment, recognize diverse risks, and provide targets with meaningful avenues for protection and redress.

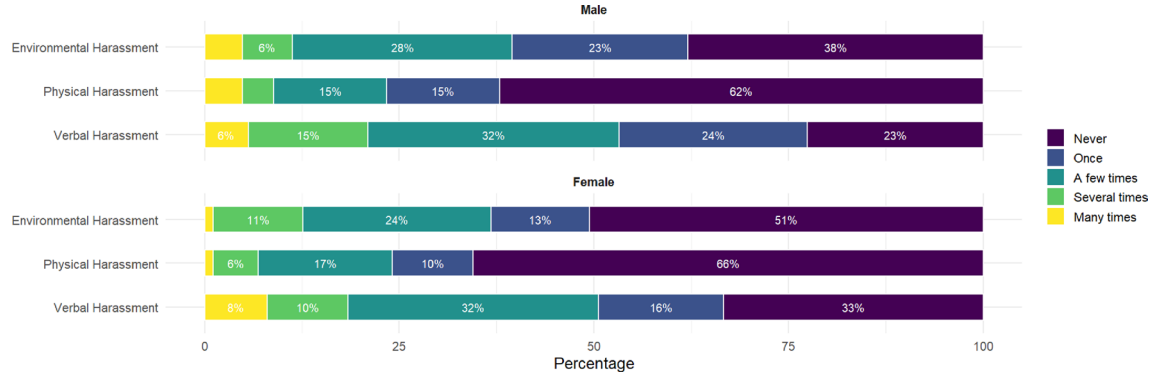
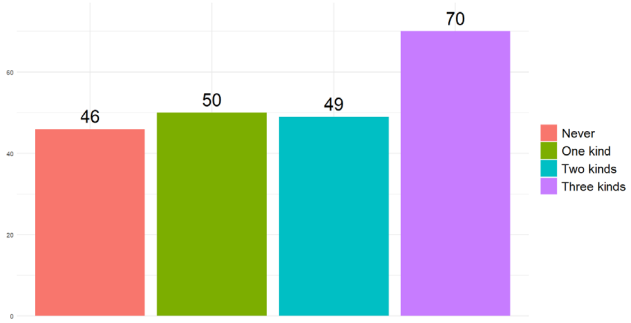
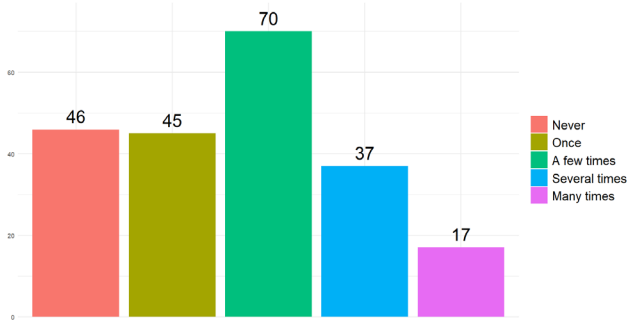


Fig. 8 Distribution of harassment frequency by gender and type



(a) Number of participants by number of harassment types experienced (verbal, physical, environmental).



(b) Maximum level of harassment experienced across all three types.

Fig. 9 Participant distribution by **a** number of harassment types experienced and **b** maximum severity level experienced in VR environments

Table 4 Gender comparison of harassment frequency by type

Harassment type	Male (M, SD)	Female (M, SD)	U	p-value
Verbal	2.57 (1.16)	2.44 (1.27)	5785	$p = 1.000$ ns
Physical	1.75 (1.14)	1.67 (1.03)	5561	$p = 1.000$ ns
Environmental	2.18 (1.16)	2.00 (1.15)	5880	$p = .720$ ns

Adjusted p-values are reported (Bonferroni correction). ns = non-significant. U = Mann–Whitney test statistic. Scale: 1 = Never, 2 = Once, 3 = A few times, 4 = Several times, 5 = Many times

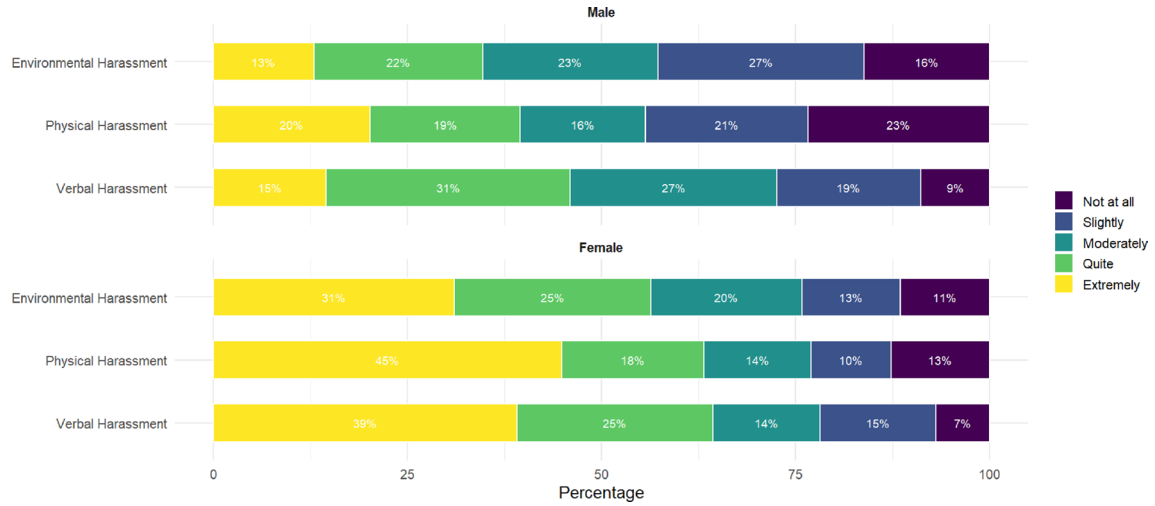


Fig. 10 Perceived harmfulness of harassment by gender and type

Table 5 Gender comparison of perceived harm by harassment type

Harassment type	Male (M, SD)	Female (M, SD)	U	p-value
Verbal	3.24 (1.18)	3.75 (1.30)	4044	$p < .005$
Physical	2.92 (1.47)	3.72 (1.44)	3730	$p < .001$
Environmental	2.89 (1.28)	3.52 (1.35)	3936	$p < .005$

Note: Adjusted p-values are reported (Bonferroni correction). U = Mann–Whitney test statistic. Scale: 1 = Not harmful at all, 2 = Slightly harmful, 3 = Moderately harmful, 4 = Quite harmful, 5 = Extremely harmful

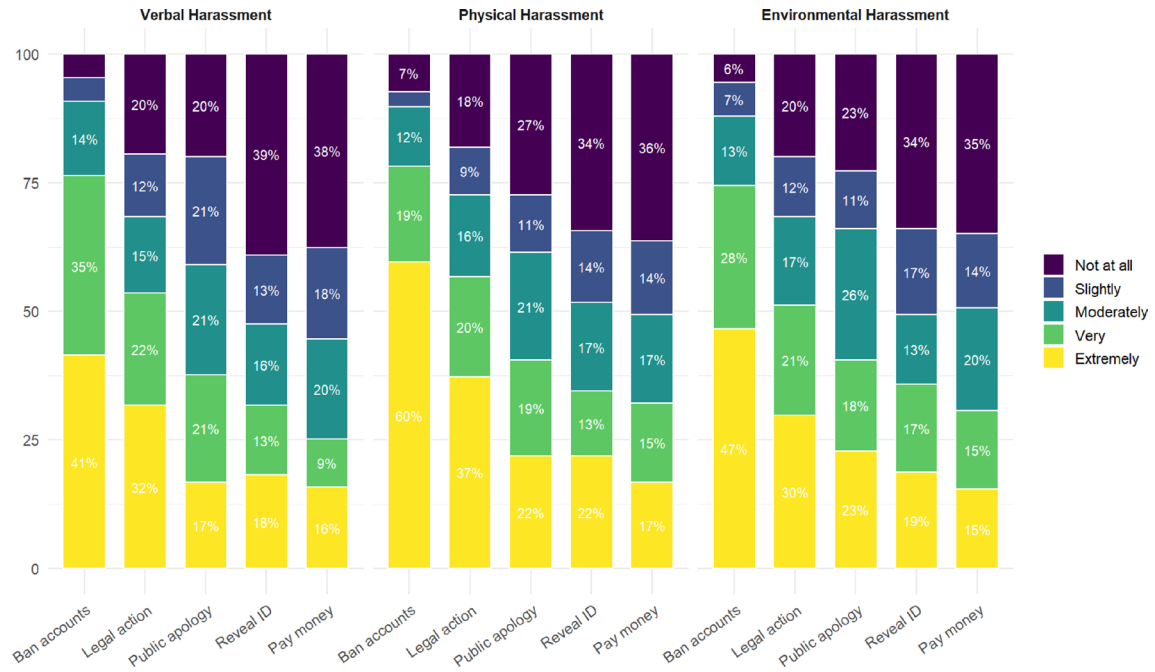


Fig. 11 Distribution of desirability ratings for response strategies across harassment types

Table 6 Post-hoc Wilcoxon signed-rank tests of desirability across response strategies

Comparison	p-value	r
Ban versus Legal Action	$p < .001$	.48
Ban versus Pay Money	$p < .001$	.36
Ban versus Reveal ID	$p < .001$	.25
Ban versus Apology	$p < .001$	.79
Legal Action versus Pay Money	$p < .001$	.08
Legal Action versus Reveal ID	$p < .001$	.19
Legal Action versus Apology	$p < .001$	.61
Apology versus Pay Money	$p < .001$	.27
Apology versus Reveal ID	$p = .002$	.34
Pay Money versus Reveal ID	$p = 1.000$ ns	.46

Bonferroni-adjusted p-values are reported. ns = non-significant. Effect size r is reported where available; r for Ban versus Legal Action was not reported in the original summary. Omnibus Friedman test: $\chi^2(4) = 228.00, p < .001$ (Kendall's $W = .27$)

Table 7 Post-hoc comparisons of response strategies across harassment types

Comparison	Verbal	Physical	Environmental
Ban versus Legal Action	$p < .001$***, $r = .438$	$p < .001$***, $r = .449$	$p < .001$***, $r = .430$
Ban versus Pay Money	$p < .001$***, $r = .696$	$p < .001$***, $r = .702$	$p < .001$***, $r = .685$
Ban versus Reveal ID	$p < .001$***, $r = .655$	$p < .001$***, $r = .662$	$p < .001$***, $r = .641$
Ban versus Apology	$p < .001$***, $r = .595$	$p < .001$***, $r = .602$	$p < .001$***, $r = .580$
Legal Action versus Pay Money	$p < .001$***, $r = .471$	$p < .001$***, $r = .479$	$p < .001$***, $r = .459$
Legal Action versus Reveal ID	$p < .001$***, $r = .478$	$p < .001$***, $r = .485$	$p < .001$***, $r = .465$
Legal Action versus Apology	$p = .002$**, $r = .266$	$p < .001$***, $r = .279$	$p = .084$, $r = .258$
Apology versus Pay Money	$p < .001$***, $r = .283$	$p < .001$***, $r = .294$	$p < .001$***, $r = .274$
Apology versus Reveal ID	$p < .001$***, $r = .246$	$p = .209$ ns, $r = .251$	$p = .050$*, $r = .239$
Pay Money versus Reveal ID	$p = .867$ ns, $r = .064$	$p = .430$ ns, $r = .071$	$p = .192$ ns, $r = .068$

*$p < .05$, **$p < .01$, ***$p < .001$. Adjusted p-values are reported; ns = non-significant

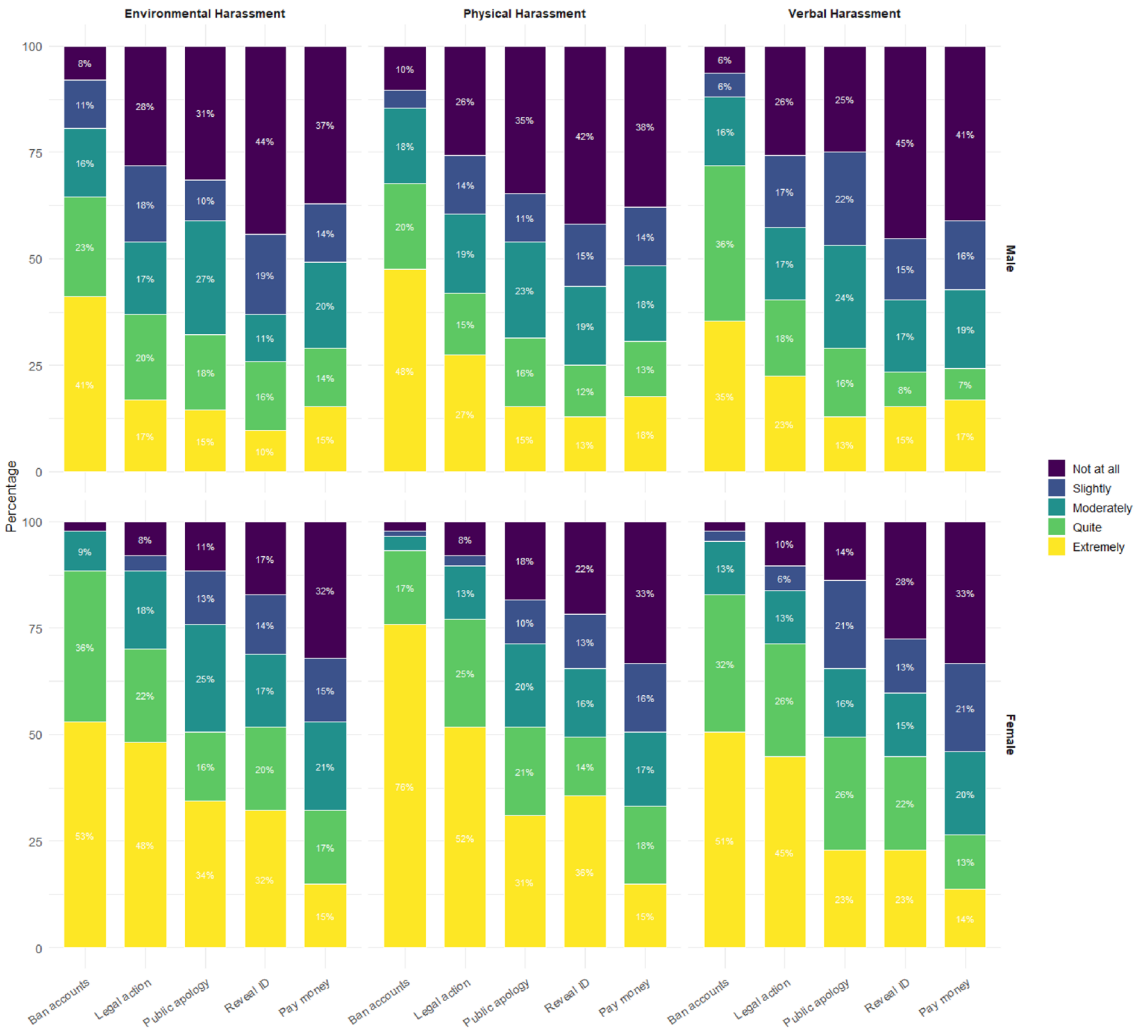


Fig. 12 Distribution of desirability ratings for response strategies by gender and harassment type

Table 8 Gender comparison of desirability ratings for response strategies (all harassment types combined)

Strategy	Male (M, SD)	Female (M, SD)	U	p-value	r
Apology	2.70 (1.42)	3.36 (1.41)	35991	$p < .001$***	0.22
Ban	3.86 (1.26)	4.42 (0.87)	36442	$p < .001$***	0.21
Legal Action	2.93 (1.51)	4.00 (1.26)	29328	$p < .001$***	0.34
Pay Money	2.53 (1.50)	2.62 (1.45)	46459	$p = .341$ ns	0.04
Reveal ID	2.34 (1.45)	3.21 (1.54)	33226	$p < .001$***	0.27

*$p < .05$, **$p < .01$, ***$p < .001$. Adjusted p-values reported (Bonferroni); ns = non-significant. U = Mann–Whitney test statistic. Scale: 1 = Not at all desirable, 2 = Slightly desirable, 3 = Moderately desirable, 4 = Very desirable, 5 = Extremely desirable

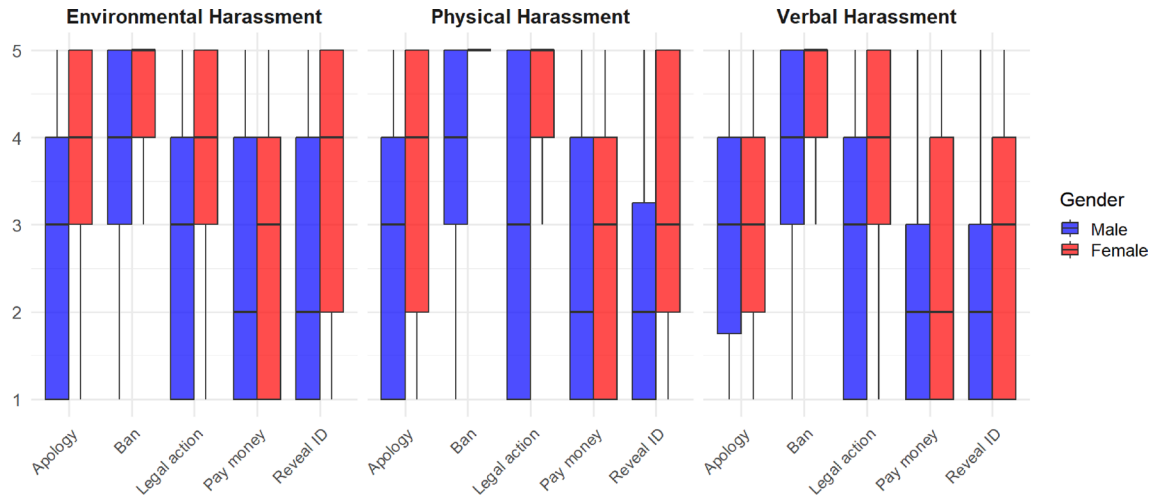


Fig. 13 Gender differences in desirability ratings across harassment types and response strategies. Female participants rated most strategies as more desirable than male participants

Table 9 Gender comparison of strategy ratings by harassment type

Harassment type	Strategy	Male (M, SD)	Female (M, SD)	U	p-value	r
Environmental	Apology	2.70 (1.44)	3.49 (1.38)	3833	$p < .001$***	.25
	Ban	3.78 (1.31)	4.37 (0.84)	4162	$p = .003$**	.19
	Legal Action	2.80 (1.47)	3.99 (1.24)	2961	$p < .001$***	.38
	Pay Money	2.56 (1.48)	2.68 (1.46)	5140	$p = .550$ ns	.04
	Reveal ID	2.28 (1.42)	3.36 (1.49)	3264	$p < .001$***	.34
Physical	Apology	2.66 (1.48)	3.36 (1.48)	3986	$p < .001$***	.22
	Ban	3.90 (1.33)	4.63 (0.81)	3648	$p < .001$***	.28
	Legal Action	3.04 (1.56)	4.10 (1.21)	3332	$p < .001$***	.33
	Pay Money	2.59 (1.53)	2.66 (1.48)	5232	$p = .701$ ns	.03
	Reveal ID	2.40 (1.45)	3.29 (1.58)	3680	$p < .001$***	.27
Verbal	Apology	2.70 (1.35)	3.24 (1.38)	4205	$p = .005$**	.19
	Ban	3.89 (1.15)	4.26 (0.93)	4384	$p = .014$*	.16
	Legal Action	2.94 (1.52)	3.90 (1.32)	3488	$p < .001$***	.30
	Pay Money	2.43 (1.50)	2.53 (1.42)	5102	$p = .489$ ns	.05
	Reveal ID	2.34 (1.49)	3.00 (1.55)	4112	$p = .002$**	.20

*$p < .05$, **$p < .01$, ***$p < .001$. Adjusted p-values reported; ns = non-significant. U = Mann–Whitney test statistic. Scale: 1 = Not at all desirable, 2 = Slightly desirable, 3 = Moderately desirable, 4 = Very desirable, 5 = Extremely desirable

Table 10 Friedman tests of response-strategy desirability within gender groups

Gender	Harassment type	$\chi^2(4)$	p-value
Male	Verbal Harassment	108.0	$< .001$
	Physical Harassment	103.0	$< .001$
	Environmental Harassment	93.1	$< .001$
Female	Verbal Harassment	93.9	$< .001$
	Physical Harassment	123.0	$< .001$
	Environmental Harassment	94.7	$< .001$

All tests were significant at $p < .001$.

Table 11 Within-gender post-hoc Wilcoxon signed-rank tests (paired; Bonferroni-adjusted) of response-strategy desirability, ordered to match the appendix figure. M_1/M_2 are strategy means (with SD) within each gender × type; reported p values are Bonferroni-adjusted; r is the effect size. Scale: 1 = Not at all desirable, 2 = Slightly desirable, 3 = Moderately desirable, 4 = Very desirable, 5 = Extremely desirable

Gender	Harassment tpe	Comparison	M_1 (SD)	M_2 (SD)	p-value	r
Male						
	Verbal	Ban versus Legal Action	3.89 (1.15)	2.94 (1.52)	$p < .001^{***}$	.53
		Ban versus Pay Money	3.89 (1.15)	2.43 (1.50)	$p < .001^{***}$	.68
		Ban versus Reveal ID	3.89 (1.15)	2.34 (1.49)	$p < .001^{***}$	.69
		Ban versus Apology	3.89 (1.15)	2.70 (1.35)	$p < .001^{***}$	.62
		Legal Action versus Pay Money	2.94 (1.52)	2.43 (1.50)	$p = .002^{**}$	.33
		Legal Action versus Reveal ID	2.94 (1.52)	2.34 (1.49)	$p < .001^{***}$	.42
		Legal Action versus Apology	2.94 (1.52)	2.70 (1.35)	$p = .629$ ns	.15
		Apology versus Pay Money	2.70 (1.35)	2.43 (1.50)	$p = .342$ ns	.24
		Apology versus Reveal ID	2.70 (1.35)	2.34 (1.49)	$p = .029^{*}$	.26
		Pay Money versus Reveal ID	2.43 (1.50)	2.34 (1.49)	$p = 1.000$ ns	.05
	Physical	Ban versus Legal Action	3.90 (1.33)	3.04 (1.56)	$p < .001^{***}$	.48
		Ban versus Pay Money	3.90 (1.33)	2.59 (1.53)	$p < .001^{***}$	.62
		Ban versus Reveal ID	3.90 (1.33)	2.40 (1.45)	$p < .001^{***}$	.69
		Ban versus Apology	3.90 (1.33)	2.66 (1.48)	$p < .001^{***}$	.62
		Legal Action versus Pay Money	3.04 (1.56)	2.59 (1.53)	$p = .010^{*}$	.33
		Legal Action versus Reveal ID	3.04 (1.56)	2.40 (1.45)	$p < .001^{***}$	.43
		Legal Action versus Apology	3.04 (1.56)	2.66 (1.48)	$p = .118$ ns	.25
		Apology versus Pay Money	2.66 (1.48)	2.59 (1.53)	$p = 1.000$ ns	.06
		Apology versus Reveal ID	2.66 (1.48)	2.40 (1.45)	$p = .216$ ns	.24
		Pay Money versus Reveal ID	2.59 (1.53)	2.40 (1.45)	$p = 1.000$ ns	.10
	Environmental	Ban versus Legal Action	3.78 (1.31)	2.80 (1.47)	$p < .001^{***}$	.54
		Ban versus Pay Money	3.78 (1.31)	2.56 (1.48)	$p < .001^{***}$	.59
		Ban versus Reveal ID	3.78 (1.31)	2.28 (1.42)	$p < .001^{***}$	.68
		Ban versus Apology	3.78 (1.31)	2.74 (1.44)	$p < .001^{***}$	.57
		Legal Action versus Pay Money	2.80 (1.47)	2.56 (1.48)	$p = .407$ ns	.20
		Legal Action versus Reveal ID	2.80 (1.47)	2.28 (1.42)	$p < .001^{***}$	.37
		Legal Action versus Apology	2.80 (1.47)	2.74 (1.44)	$p = 1.000$ ns	.07
		Apology versus Pay Money	2.74 (1.44)	2.56 (1.48)	$p = 1.000$ ns	.15
		Apology versus Reveal ID	2.74 (1.44)	2.28 (1.42)	$p = .001^{***}$	.32
		Pay Money versus Reveal ID	2.56 (1.48)	2.28 (1.42)	$p = .347$ ns	.19
Female						
	Verbal	Ban versus Legal Action	4.26 (0.93)	3.90 (1.32)	$p = .108$ ns	.25
		Ban versus Pay Money	4.26 (0.93)	2.53 (1.42)	$p < .001^{***}$	.74
		Ban versus Reveal ID	4.26 (0.93)	3.00 (1.55)	$p < .001^{***}$	.62
		Ban versus Apology	4.26 (0.93)	3.24 (1.38)	$p < .001^{***}$	.61
		Legal Action versus Pay Money	3.90 (1.32)	2.53 (1.42)	$p < .001^{***}$	.65
		Legal Action versus Reveal ID	3.90 (1.32)	3.00 (1.55)	$p < .001^{***}$	.55
		Legal Action versus Apology	3.90 (1.32)	3.24 (1.38)	$p = .002^{**}$	.36
		Apology versus Pay Money	3.24 (1.38)	2.53 (1.42)	$p = .002^{**}$	.44
		Apology versus Reveal ID	3.24 (1.38)	3.00 (1.55)	$p = 1.000$ ns	.18
		Pay Money versus Reveal ID	2.53 (1.42)	3.00 (1.55)	$p = .159$ ns	.22
	Physical	Ban versus Legal Action	4.63 (0.81)	4.10 (1.21)	$p < .001^{***}$	.415
		Ban versus Pay Money	4.63 (0.81)	2.66 (1.48)	$p < .001^{***}$	.803
		Ban versus Reveal ID	4.63 (0.81)	3.29 (1.58)	$p < .001^{***}$	.711
		Ban versus Apology	4.63 (0.81)	3.36 (1.48)	$p < .001^{***}$	.711

Table 11 (continued)

Gender	Harassment tpe	Comparison	M_1 (SD)	M_2 (SD)	p-value	r
		Legal Action versus Pay Money	4.10 (1.21)	2.66 (1.48)	$p < .001^{***}$	.714
		Legal Action versus Reveal ID	4.10 (1.21)	3.29 (1.58)	$p < .001^{***}$	.555
		Legal Action versus Apology	4.10 (1.21)	3.36 (1.48)	$p < .001^{***}$	.462
		Apology versus Pay Money	3.36 (1.48)	2.66 (1.48)	$p = .004^{**}$	.413
		Apology versus Reveal ID	3.36 (1.48)	3.29 (1.58)	$p = 1.000$ ns	.048
		Pay Money versus Reveal ID	2.66 (1.48)	3.29 (1.58)	$p = .016^{*}$	.302
	Environmental	Ban versus Legal Action	4.37 (0.84)	3.99 (1.24)	$p = .018^{*}$	.288
		Ban versus Pay Money	4.37 (0.84)	2.68 (1.46)	$p < .001^{***}$	.764
		Ban versus Reveal ID	4.37 (0.84)	3.36 (1.49)	$p < .001^{***}$	.540
		Ban versus Apology	4.37 (0.84)	3.49 (1.38)	$p < .001^{***}$	.532
		Legal Action versus Pay Money	3.99 (1.24)	2.68 (1.46)	$p < .001^{***}$	.716
		Legal Action versus Reveal ID	3.99 (1.24)	3.36 (1.49)	$p < .001^{***}$	.525
		Legal Action versus Apology	3.99 (1.24)	3.49 (1.38)	$p = .005^{**}$	.363
		Apology versus Pay Money	3.49 (1.38)	2.68 (1.46)	$p < .001^{***}$	.500
		Apology versus Reveal ID	3.49 (1.38)	3.36 (1.49)	$p = 1.000$ ns	.105
		Pay Money versus Reveal ID	2.68 (1.46)	3.36 (1.49)	$p = .002^{**}$	.365

Appendix 1: Distribution of reported harassment

Appendix 1.1: Distribution of harassment frequency by gender and type

Verbal harassment was the most frequently experienced type across genders. Among males ($n = 124$), only 28 participants (22.6%) reported never experiencing it, while 30 (24.2%) reported it once, 40 (32.3%) a few times, 19 (15.3%) several times, and 7 (5.6%) many times. Altogether, 26 participants (21.0%) reported frequent verbal harassment (i.e., “Several times” or “Many times”). Among females ($n = 87$), 29 participants (33.3%) reported never experiencing verbal harassment, 14 (16.1%) once, 28 (32.2%) a few times, 14 (16.1%) several times, and 2 (2.3%) many times. In total, 16 participants (18.4%) experienced frequent verbal harassment. *Physical harassment* was the least reported type. For males, 77 participants (62.1%) never experienced it, 18 (14.5%) once, 18 (14.5%) a few times, 5 (4.0%) several times, and 6 (4.8%) many times. Only 11 participants (8.9%) reported frequent physical harassment. For females, 57 participants (65.5%) never experienced it, 9 (10.3%) once, 15 (17.2%) a few times, 3 (3.4%) several times, and 3 (3.4%) many times, with 6 participants (6.9%) reporting frequent occurrences. *Environmental harassment* fell between the two. For males, 47 participants (37.9%) never experienced it, 28 (22.6%) once, 35 (28.2%) a few times, 9 (7.3%) several times, and 5 (4.0%) many times, with 14 participants (11.3%) reporting frequent experiences. For females, 44 participants (50.6%) never experienced it, 11 (12.6%) once, 21 (24.1%) a few times, 10 (11.5%) several times, and 1 (1.1%) many times. Overall, 11 participants (12.6%) reported frequent environmental harassment. These gender-specific distributions are visualized in Fig. 8.

Appendix 1.2: Overview of reported harassment

Figure 9a illustrates the number of participants based on the number of harassment types they have experienced in a VR environment. The values indicate that nearly one-third (32.6%, 70 participants) of participants experienced all three types of harassment, while only 21.4% never experienced any harassment in VR. Figure 9b illustrates the maximum harassment level experienced by participants across three harassment categories (verbal, physical, and environmental harassment). More than half (57%) of participants experienced harassment at least "A few times" or more in one of the harassment categories. 27.4% of participants experienced frequent harassment ("Several times" or "Many times"). The most common harassment level is "A few times" (30.3%). 21.4% (46 participants) reported never experiencing harassment in any of the three categories.

Appendix 1.3: Gender comparison of harassment frequency by type

See Table 4.

Appendix 2: Perceived harm

Appendix 2.1: Harmfulness across harassment types

As shown in Fig. 4, *verbal harassment* was most commonly perceived as moderately or extremely harmful, with 63 participants (29.3%) and 53 participants (24.7%) selecting these categories, respectively. Only 17 participants (7.9%) considered it not harmful at all. *Physical harassment* was rated as extremely harmful by 66 participants (30.7%), while 41 (19.1%) considered it quite harmful. However, an equal number of participants (41, 19.1%) rated it as not harmful at all, indicating a polarized perception. Perceptions of *environmental harassment* were more evenly distributed: 49 participants (22.8%) rated it moderately harmful, 46 (21.4%) extremely harmful, 45 (20.9%) slightly harmful, and 44 (20.5%) quite harmful. Only 31 participants (14.4%) found it not harmful at all.

Appendix 2.2: Harmfulness of harassment by gender and type

Verbal harassment was perceived as the most harmful overall. Among males ($n = 124$), 11 participants (8.9%) reported "Not at all" harmful, 23 (18.5%) "Slightly," 33 (26.6%) "Moderately," 39 (31.5%) "Quite," and 18 (14.5%) "Extremely." Altogether, 57 participants (46.0%) perceived verbal harassment as highly harmful (i.e., "Quite" or "Extremely"). The mean score was 3.24 (median = 3). Among females ($n = 87$), only 4 participants (4.6%) reported "Not at all," while 9 (10.3%) rated it "Slightly," 18 (20.7%) "Moderately," 29 (33.3%) "Quite," and 27 (31.0%) "Extremely." In total, 56 participants (64.4%) reported high perceived harmfulness. The mean score was 3.75 (median = 4). *Physical harassment* was rated slightly less harmful by men but similarly high by women. For males, 29 (23.4%) reported "Not at all," 26 (21.0%) "Slightly," 20 (16.1%) "Moderately," 24 (19.4%) "Quite," and 25 (20.2%) "Extremely." Overall, 49 participants (39.5%) perceived high harmfulness (mean = 2.92, median = 3). For females, 10 (11.5%) reported "Not at all," 9 (10.3%) "Slightly," 13 (14.9%) "Moderately," 24 (27.6%) "Quite," and 31 (35.6%) "Extremely." Altogether, 55 participants (63.2%) perceived high harmfulness (mean = 3.72, median = 4). *Environmental harassment* fell in between. Among males, 28 (22.6%) reported "Not at all," 26 (21.0%) "Slightly," 27 (21.8%) "Moderately," 23 (18.5%) "Quite," and 20 (16.1%) "Extremely." A total of 43 participants (34.7%) perceived high harmfulness (mean = 2.89, median = 3). Among females, 15 (17.2%) reported "Not at all," 11 (12.6%) "Slightly," 12 (13.8%) "Moderately," 26 (29.9%) "Quite," and 23 (26.4%) "Extremely." Altogether, 49 participants (56.3%) perceived high harmfulness (mean = 3.52, median = 4). These gender-specific distributions of perceived harmfulness are visualized in Fig. 10.

Appendix 2.3: Gender comparison of perceived harm by harassment type

See Table 5.

Appendix 3: Comparison of strategy ratings

Appendix 3.1: Perceived desirability of response strategies across different harassment types

Across all harassment types, *ban accounts* was consistently rated as the most desirable intervention. For *verbal harassment* ($n = 215$ per strategy), 75 participants (34.9%) rated banning accounts as "Very" desirable and 89 (41.4%) as "Extremely," amounting to 76.3% in the high-desirability categories. Legal action followed, with 115 participants (53.5%) in high-desirability. Public apology received balanced evaluations: 36 participants (16.7%) rated it "Extremely," 45 (20.9%) "Very," while 43 (20.0%) selected "Not at all." Reveal ID and pay money were less supported, with only 31.6% and 25.1% respectively in high-desirability. For *physical harassment*, banning accounts again dominated: 168 participants (78.1%) rated it as highly desirable. Legal action also attracted broad support (122 participants; 56.7% high-desirability). Public apology and reveal ID drew more mixed views (40.5% and 34.5% high-desirability, respectively). Pay money remained the least endorsed (32.0%). For *environmental harassment*, banning accounts reached the highest endorsement overall: 160 participants (74.4%) rated it as highly desirable. Legal action was next (51.2%), followed by public apology (40.5%), reveal ID (35.8%), and pay money (30.6%). In summary, *ban accounts* consistently emerged as the most strongly supported response strategy across harassment types, while *pay money* was the least favored. Legal action occupied an intermediate position, whereas public apology and reveal ID received mixed evaluations, see Fig. 11 for details.

Appendix 3.2: Post-hoc Wilcoxon signed-rank tests of desirability across response strategies

See Table 6.

Appendix 3.3: Post-hoc comparisons of response strategies across harassment types

See Table 7.

Appendix 3.4: Perceived desirability of response strategies by gender

Across genders, *ban accounts* consistently emerged as the most desirable intervention, while *pay money* was the least favored. For *verbal harassment*, male participants ($n = 124$) showed strong support for banning accounts, with 45 (36.3%) rating it "Very" and 44 (35.5%) "Extremely." Legal action was endorsed less strongly, with 22 (17.7%) rating it "Very" and 28 (22.6%) "Extremely." Public apology was more evenly spread, with only 16 (12.9%) rating it "Extremely," while 31 (25.0%) reported "Not at all." Reveal ID and pay money were least endorsed, with over 40% reporting "Not at all." Among females ($n = 87$), endorsement of banning accounts was even stronger: 28 (32.2%) rated it "Very" and 44 (50.6%) "Extremely." Legal action was also highly valued (23; 26.4% "Very" and 39; 44.8% "Extremely"). Public apology received more mixed ratings, with 20 (23.0%) "Extremely," while 18 (20.7%) reported "Slightly." Reveal ID attracted lower support (23.0% "Extremely"), and pay money remained least desirable (only 13.8% "Extremely"). For *physical harassment*, males most strongly favored banning accounts (59; 47.6% "Extremely"), while legal action received moderate support (34; 27.4% "Extremely"). Public apology, reveal ID, and pay money were rated lower, each with less than 20% in "Extremely." In contrast, female participants overwhelmingly endorsed both banning accounts (66; 75.9% "Extremely") and legal action (45; 51.7% "Extremely"). Public apology and reveal ID were moderately supported (31.0% and 35.6% "Extremely," respectively), while pay money remained the least supported (14.9% "Extremely"). For *environmental harassment*, males again showed strongest endorsement for banning accounts (51; 41.1% "Extremely"), with legal action and public apology both below 20% "Extremely." Reveal ID and pay money had limited support (9.7–15.3% "Extremely"). Female participants strongly endorsed banning accounts (46; 52.9% "Extremely") and legal action (42; 48.3% "Extremely"). Public apology and reveal ID were moderately rated (34.5% and 32.2% "Extremely," respectively), while pay money attracted little support (14.9% "Extremely"). In summary, banning accounts was the most consistently endorsed response strategy across genders and harassment types, with particularly strong female support. Legal action was also broadly supported, especially by females. Public apology and reveal ID received mixed ratings, and pay money was consistently least desirable. These gender-specific distributions are visualized in Fig. 12.

Appendix 3.5: Gender comparison of desirability ratings for response strategies (all harassment types combined)

See Table 8.

Appendix 3.6: Gender differences in desirability ratings across harassment types and response strategies

See Fig. 13.

Appendix 3.7: Gender comparison of strategy ratings by harassment type

See Table 9.

Within-gender comparisons of response-strategy desirability

See Table 10 and 11.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10055-026-01388-1.

Author contributions Qijia Chen collected the data, performed the analyses, and drafted the manuscript. Andrea Bellucci contributed to data analysis, manuscript writing, and revision. Qunfang Wu helped structure the article and revised the text. Giulio Jacucci provided overall guidance and feedback and contributed to revising the manuscript.

Funding Open Access funding provided by University of Helsinki (including Helsinki University Central Hospital). This research was funded by the Research Council of Finland (#357270).

Data availability Data are available upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright

holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Bartlett J, Norrie R, Patel S, Rumpel R, Wibberley S (2014) Misogyny on twitter. Demos 1–18

Bekiari A, Hassanagas N (2015) Verbal aggressiveness exploration through complete social network analysis: using physical education students' class as an illustration. Int'l J Soc Sci Stud 3:30

Blackwell L, Dimond J, Schoenebeck S, Lampe C (2017) Classification and its consequences for online harassment: design insights from heartmob. In: Proceedings of the ACM on human-computer interaction 1(CSCW), 1–19

Blackwell L, Ellison N, Elliott-Deflo N, Schwartz R (2019) Harassment in social virtual reality: challenges for platform governance. In: Proceedings of the ACM on human-computer interaction 3(CSCW), 1–25

Boellstorff T (2015) Coming of age in second life: an anthropologist explores the virtually Human. Princeton University Press, USA

Bruckman A, Curtis P, Figallo C, Laurel B (1994) Approaches to managing deviant behavior in virtual communities. In: Conference companion on human factors in computing systems. CHI '94, pp. 183–184. Association for Computing Machinery, New York https://doi.org/10.1145/259963.260231

Celuch M, Oksa R, Ellonen N, Oksanen A (2024) Self-censorship among online harassment targets: the role of support at work, harassment characteristics, and the target's public visibility. Inf Commun Soc 27(11):2171–2190

Chen Q, Spapé MM, Jacucci G (2024) Understanding phantom tactile sensation on commercially available social virtual reality platforms. ACM Hum Comput Interact Proc. https://doi.org/10.1145/3637418

Chen Q, Cai J, Jacucci G (2024) "People are way too obsessed with rank": Trust system in social virtual reality. Comput Support Cooperat Work (CSCW) 33(4):925–957

Chen Q, Bellucci A, Cai J, Nelimarkka M, Jacucci G (2025) Understanding "mutes" in social virtual reality. In: Proceedings of the 2025 CHI conference on human factors in computing systems. CHI '25. Association for Computing Machinery, New York https://doi.org/10.1145/3706598.3714244

Citron DK (2014) Hate crimes in cyberspace. Harvard University Press, Cambridge

Citron DK (2017) Extremist speech, compelled conformity, and censorship creep. Notre Dame L Rev 93:1035

Filatova-Bilous N (2021) Once again platform liability: on the edge of the'uber'and'airbnb'cases

Fox J, Tang WY (2017) Women's experiences with general and sexual harassment in online video games: rumination, organizational responsiveness, withdrawal, and coping strategies. New Media Society 19(8):1290–1307

Freeman G, Acena D (2022) " Acting out" queer identity: the embodied visibility in social virtual reality. Proc ACM Hum-Comput Interact 6(CSCW2):1–32

Freeman G, Maloney D (2021) Body, avatar, and me: The presentation and perception of self in social virtual reality. Proc ACM Hum-Comput Interact 4(CSCW3):1–27

Freeman G, Zamanifard S, Maloney D, Acena D (2022) Disturbing the peace: experiencing and mitigating emerging harassment in social virtual reality. Proc ACM Hum-Comput Interact 6(CSCW1):1–30

Gray KL (2014) Race, gender, and deviance in xbox live: theoretical perspectives from the virtual margins. Routledge, London

Im J, Schoenebeck S, Iriarte M, Grill G, Wilkinson D, Batool A, Alharbi R, Funwie A, Gankhuu T, Gilbert E, Naseem M (2022) Women's perspectives on harm and justice after online harassment. Proc ACM Hum-Comput Interact 6:1–23. https://doi.org/10.1145/3555775

Im J, Schoenebeck S, Iriarte M, Grill G, Wilkinson D, Batool A, Alharbi R, Funwie A, Gankhuu T, Gilbert E et al (2022) Women's perspectives on harm and justice after online harassment. Proc ACM Hum-Comput Interact 6(CSCW2):1–23

Jane EA (2014) "Your a ugly, whorish, slut" understanding e-bile. Fem Media Stud 14(4):531–546

Jiang JA, Scheuerman MK, Fiesler C, Brubaker JR (2021) Understanding international perceptions of the severity of harmful content online. PLoS ONE 16(8):0256762

Kilteni K, Groten R, Slater M (2012) The sense of embodiment in virtual reality. Presence Teleop Virt 21(4):373–387

Kou Y (2021) Punishment and its discontents: an analysis of permanent ban in an online game community. Proc ACM Hum-Comput Interact 5(CSCW2):1–21

Lenhart A, Ybarra M, Zickuhr K, Price-Feeney M (2016) Online harassment, digital abues, and cyberstalking in America

Li L, Freeman G, Schulenberg K, Acena D (2023) " We cried on each other's shoulders": How lgbtq+ individuals experience social support in social virtual reality. In: Proceedings of the 2023 CHI conference on human factors in computing systems, pp. 1–16

Ma R, You Y, Gui X, Kou Y (2023) How do users experience moderation?: A systematic literature review. Proc ACM Hum-Comput Interact 7(CSCW2):1–30

Ma R, Li Y, Kou Y (2023) Transparency, fairness, and coping: How players experience moderation in multiplayer online games. In: Proceedings of the 2023 CHI conference on human factors in computing systems, pp. 1–21

Maloney D, Freeman G, Wohn DY (2020) "Talking without a voice" understanding non-verbal communication in social virtual reality. Proc ACM Hum-Comput Interact 4(CSCW2):1–25

Maloney D, Freeman G (2020) Falling asleep together: what makes activities in social virtual reality meaningful to users. In: Proceedings of the annual symposium on computer-human interaction in play, pp. 510–521

Maloney D, Freeman G, Robb A (2021) Social virtual reality: ethical considerations and future directions for an emerging research space. In: 2021 IEEE conference on virtual reality and 3D user interfaces abstracts and workshops (VRW), pp. 271–277. IEEE

McGlynn C, Rigotti C (2025) From virtual rape to meta-rape: sexual violence, criminal law and the metaverse. Oxf J Leg Stud 45(3):554–582. https://doi.org/10.1093/ojls/gqaf009

McVeigh-Schultz J, Kolesnichenko A, Isbister K (2019) Shaping pro-social interaction in vr: an emerging design framework. In: Proceedings of the 2019 CHI conference on human factors in computing systems, pp. 1–12

McVeigh-Schultz J, Márquez Segura E, Merrill N, Isbister K (2018) What's it mean to" be social" in vr? mapping the social vr design ecology. In: Proceedings of the 2018 ACM conference companion publication on designing interactive systems, pp. 289–294

Piitulainen R, Hämäläinen P, Mekler ED (2022) Vibing together: dance experiences in social virtual reality. In: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems, pp. 1–18

Pittaro ML (2007) Cyber stalking: an analysis of online harassment and intimidation. Int J Cyber Criminol 1(2):180–197

Quayle E (2020) Prevention, disruption and deterrence of online child sexual exploitation and abuse. In: Era Forum, vol. 21, pp. 429–447. Springer

Richardson M, Cork AG, Stanton Fraser D, Proulx MJ, Pan X, Krauß V, Khamis M, Lukosch H (2024) Shaping the future: developing principles for policy recommendations for responsible innovation in virtual worlds. In: Extended abstracts of the CHI conference on human factors in computing systems, pp. 1–6

Sabri N, Chen B, Teoh A, Dow SP, Vaccaro K, Elsherief M (2023) Challenges of moderating social virtual reality. In: Proceedings of the 2023 CHI conference on human factors in computing systems, pp. 1–20

Sambasivan N, Batool A, Ahmed N, Matthews T, Thomas K, Gaytán-Lugo LS, Nemer D, Bursztein E, Churchill E, Consolvo S (2019) " They don't leave us alone anywhere we go" gender and digital abuse in South Asia. In: Proceedings of the 2019 CHI conference on human factors in computing systems, pp. 1–14

Schoenebeck S, Haimson OL, Nakamura L (2021) Drawing from justice theories to support targets of online harassment. New Media Society 23(5):1278–1300

Schoenebeck S, Lampe C, Trieu P (2023) Online harassment: assessing harms and remedies. Soc Media Soc. https://doi.org/10.1177/20563051231157297

Schoenebeck S, Batool A, Do G, Darling S, Grill G, Wilkinson D, Khan M, Toyama K, Ashwell L (2023) Online harassment in majority contexts: examining harms and remedies across countries. In: Proceedings of the 2023 CHI conference on human factors in computing systems, pp. 1–16

Schulenberg K, Freeman G, Li L, Barwulor C (2023) " Creepy towards my avatar body, creepy towards my body": how women experience and manage harassment risks in social virtual reality. Proc ACM Hum-Comput Interact 7(CSCW2):1–29

Schulenberg K, Freeman G, Li L, Knijnenburg BP (2024) Does who you are or appear to be matter?: Understanding identity-based harassment in social vr through the lens of (mis) perceived identity revelation. Proc ACM Hum-Comput Interact 8(CSCW2):1–40

Slater M, Sanchez-Vives MV (2016) Enhancing our lives with immersive virtual reality. Front Robot AI 3:236866

Stanaland A, Gaither S (2020) "Be a man": the role of social pressure in eliciting men's aggressive cognition. Pers Soc Psychol Bull 47:1596–1611. https://doi.org/10.1177/0146167220984298

Stevens F, Nurse JR, Arief B (2021) Cyber stalking, cyber harassment, and adult mental health: a systematic review. Cyberpsychol Behav Soc Netw 24(6):367–376

Strong DR, Leas E, Elton-Marshall T, Wackowski OA, Travers M, Bansal-Travers M, Hyland A, White M, Noble M, Cummings KM et al (2019) Harm perceptions and tobacco use initiation among youth in wave 1 and 2 of the population assessment of tobacco and health (path) study. Prev Med 123:185–191

Sultana S, Guimbretière F, Sengers P, Dell N (2018) Design within a patriarchal society: Opportunities and challenges in designing for rural women in Bangladesh. In: Proceedings of the 2018 CHI conference on human factors in computing systems, pp. 1–13

Sykownik P, Graf L, Zils C, Masuch M (2021) The most social platform ever? A survey about activities & motives of social vr users. In: 2021 IEEE virtual reality and 3D user interfaces (VR), pp. 546–554. IEEE

Tanenbaum TJ, Hartoonian N, Bryan J (2020) " How do i make this thing smile?" an inventory of expressive nonverbal communication in commercial social virtual reality platforms. In: Proceedings of the 2020 CHI conference on human factors in computing systems, pp. 1–13

Thomas K, Akhawe D, Bailey M, Boneh D, Bursztein E, Consolvo S, Dell N, Durumeric Z, Kelley PG, Kumar D et al (2021) Sok: Hate, harassment, and the changing landscape of online abuse. In: 2021 IEEE symposium on security and privacy (SP), pp. 247–267. IEEE

Tyler TR (2003) Procedural justice, legitimacy, and the effective rule of law. Crime Just 30:283–357

Ventura S, Cardenas G, Miragall M, Riva G, Baños R (2021) How does it feel to be a woman victim of sexual harassment? The effect of 360-video-based virtual reality on empathy and related variables. Cyberpsychol Behav Soc Netw 24(4):258–266

Vogels EA (2021) The state of online harassment. Accessed 02 July 2025. https://www.pewresearch.org/internet/2021/01/13/the-state-of-online-harassment/

Weerasinghe M, Macdonald S, Fiani C, O'Hagan J, Chollet M, McGill M, Khamis M (2025) Beyond mute and block: adoption and effectiveness of safety tools in social vr, from ubiquitous harassment to social sculpting. IEEE Trans Visual Comput Gr. https://doi.org/10.1109/TVCG.2025.3549860

Ybarra ML (2004) Linkages between depressive symptomatology and internet harassment among young regular internet users. Cyberpsychol Behav 7(2):247–257

Ybarra ML, Mitchell KJ (2007) Prevalence and frequency of internet harassment instigation: implications for adolescent health. J Adolesc Health 41(2):189–195

Zehr H (2015) The little book of restorative justice: revised and updated. Simon and Schuster, New York

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.